Yeah, maybe that he's just doing it because I'm away in the US is managing the facial period that we've got to go over here.
It's just past nine minutes. Oh, yeah, yeah.
Oh Nicole, I'll let you start whenever you need to talk about something.
How many are we waiting for? Two way away on two people.
We won't wait on other mornings, so if you think that we're gonna wait for you every morning, we're not.
Can I give you this? Are you okay just to morning everyone? Morning. Good morning, everyone. Um mayor may not know me on the policeman the program coordinator.
Um, before we get started, and I give the floor to Michael and Sophie for a quick intro. Um I'm just gonna play this video from leadership. Um, just a couple of housekeeping items.
We do have voucher code for lunch the entire week. That's your code. It's also in the chat if you forget. Um, and then if you have the occasional conflict with the meeting or something, you can step out as long as it's not really disruptive, that's fine.
Um just let us know if you're gonna be belong gone for a long time. Um we don't want you to miss too much.
We're so excited to have you here. This is an important part of capital groups investment in data excellence. You're here because we believe you have the potential to succeed in this role.
The training here is gonna help you develop the skills and experience needed to make that impact in data and our data excellence program here a capital group.
As our firm becomes increasingly data frequent, you'll help transform business needs into trusted reusable data products that help us deliver business priorities at scale.
Please treat this training as a hands-on learning experience. Throughout the week, you'll apply concepts, practice new skills, and work with scenarios that reflect what you'll actually do as a data product creator.

We don't expect that you've arrived here today, knowing everything. We expect that we show up every day, engage with materials, and work through it with the energy and positivity that you would be doing to all of it.
This program is still evolving. Your feedback matters. If something isn't clear, if you need support, if you have ideas for improvement, let us know.
And one last thing. Please stay curious. Ask questions and engage fully in the experience. We're really glad you're here. We'll look forward to seeing the impact you will make.
Michael before. Alright, good morning, everyone. Good morning. Thank you. Good morning, one, I suppose, right? Good morning. There we are. So my name is Michael Burgess.
I'm uh what shall we say? I'm the head of technical consulting at uh decoded. Can you give me my name and so on? So my name is Michael, Michael Burgess, and I'm the head of technical consulting at a company called Decoded.
And Decoded is an international skills consultancy. Uh it does many other things as well. Um and uh we work in the United States and Hong Kong and UK and other places.
Um we work with very large organizations, often in finance. Personally, I work in finance and historically I work in defense as well. And um, what do I do for decoded?
I'm the head of clinical consulting, so it's my job to work with uh clients and develop the first design of a program. Uh hopefully I then hand it over when the design is stable to other people.
Um my background is I was um so what is my background? If you want to know my background, it'd be interesting, right? We can kill some time uh rambling about me.
That sounds good, that just sounds good, right? Me rambling about me, I love that, right? Uh so I uh I had a web, uh I had a web uh startup when I was 17, so actually given a web startup by a company called Borland.
You may some of you may know what Borland was, major web consultancy in

Silicon Valley. Um what did I do after that? Then uh so I was a teenager and I was a web a CTO of Webstar, and then what did I do?
I um did physics, I was a contractor for 10 years in software engineering and defense. I worked in um for defense companies. I also work for um of the Olympics in 2012.
I worked for uh what else should I do? So many different things work for orange. My first job was uh orange contractor, contractor for orange. You know Orange Key Mobile?
You know T Mobile, right? It's the same same company. Then what? Then uh in my 20s, I took what was called retirement for me, and I started working in education, so that's where I am now.
Uh so less stressful, hopefully. And uh for the last 10 years I've been doing this. So, what is this? This is again partnering with uh often large multinational organizations and their uh SMEs and then the L and D team and giving initial uh iterations on program design.

Cool. Hi everyone, my name is Sophie McMurray. Um I've been working in data focused roles for the past nine years, and my background is in maths and statistics, and then I did a master's degree in data science.
And um at decoded, I'm working primarily on our AI engineering programs and also supporting on this program. So I'll be around to answer any questions you have.
I'll be walking around the room and yeah, looking forward to getting to know everybody. And my educational background is physics. You do medicine for a while, medicine for two years and decided it didn't like people.
Then I transferred to physics. I sat I sat outside of a physics office and said, I can't bother doing anything. What do you want to do? So they said, Do you want to do physics?
I said yes. And it was a bad decision because uh I'd have been better off doing theoretical physics, but they put me on a normal physics course and there's lots of experiments to do, and I hated it.

So I drag kicking and screaming through an experimental physics degree. Uh anyway, the reason I'm being a little bit chatter this morning is because we have Hero coming up in a couple of minutes, so I don't really want to do anything else. Let me say a little bit about the course. But once we've talked about what we've had hero across to talk about additional things, we'll get some interruptions from you and I'll introduce the course a bit more.
But one of the first things we'll do after Hero is we'll do a group exercise across your tables where you introduce each other. Um, you talk about some data products, you talk about some hobbies and some other things, icebreaker type stuff, and then we'll do a group debrief, and you can introduce yourself there. That's why I'm delaying that a little bit, so we'll do that when Hero's finish speaking. So we're waiting on Hero in a be here in a couple of minutes, I guess. Um what should we talk about then? Um here in ten minutes. Ten minutes. So we could do that channel. We could do that now then, couldn't we? Yeah.
I mean, 10 minutes is a little sharp. Usually I give people a little bit longer than that. Um so we can at least do the first part of the challenge.
We'll do the second part on heroes here. Cool. So the group challenge has two parts. The first part is just some icebreakers, introduce yourself.
And the second part is talk about some data products and pain points around data products that you're already familiar with. So we'll do that part second.
So, in terms of having you introduce introduce yourselves and each other, why don't we do? Um, first of all, go around the so here's what we're gonna do, right? Go around each group, introduce yourselves. So, this is what team you're from, right? What your role is in that team, right? Uh, any hobbies you have, and then share, want to know what the shared hobbies across the team are.
If there are if there are any, cool. And what else? Look very depressed. Some people look very depressed this morning. What did Monday? Rainybody.
Maybe we can give them until 9.30 to chat, and then Hero can join, and then they can you can do your little share out after. Sure. Okay. Okay. You can do it all together then.

You can try and squeeze it in 50 minutes. So the last thing I want you to just go ask then after your shared hobbies, is uh any existing data products you're working on.
And I want you to kind of gossip and bitch about what's wrong with them. Uh so you have um some paper on the desks and some pens. The paper and pens is mostly to do that brainstorm on data products you're working on.
So it'd be nice to know what sort of data products you've been working on across the team. And if you can do a little brainstorm on that, that'd be very helpful.
Because as we progress through the course, we'll refer back to some of those and some other additional challenges. So you have some bases to work from.
So for example, if I say, for example, use one of the things you discussed earlier, then you've got that list of things to refer back to. That makes sense.
Okay, so you've got 15 minutes to introduce yourselves, teams roles, shared hobbies, do that relatively quickly, five minutes, whatever. Then spend 10 minutes going through some existing data products you're working on and some issues that you have with them.
Cool. Begin. Go. So I'm saying I'd be doing CD.
It's all in Rosemary's head. Okay, great. Well, why don't we have a wonderful contribution from Hero? Now and when Hero's done, we'll come back and we'll do some brains on me together.
Cool? Alright. April Mummy. Good morning, everybody. Hopefully. Not what it here. I need some notes. Okay. But uh, I'm glad you're all here. Let's say there's an arrainy uh micro.
Yeah. I believe I look across the room. I know most of you, not everybody. Uh can I just get a show of hands? Who here is from this the client sales and marketing team?
Anyone? Yeah. Right. Sales marketing? Global client group. Global client group, yeah. That's how you know how long you've been at CG. I don't think you can change my name yet.

Global, what's it called? Global client group for global product. Yeah, most of you? Right. How about uh am I going to say it's right? Io tech. Investment operations.
One, alright. Investment group, IG Tech. One. Alright. Edo. You. Alright. Anyone else? Finance. Corporate. Anything? GBS? No? Okay, good. So we have a representative from IG, IO, and local client group.
And you're probably wondering, how did you get so lucky to be voluntary to join this class? For how many days? Five in total. Great. You know, before it was six days.
We cut it down to five days, so you don't have to come back in one day. Yeah. That's right. Right? Okay, good. Alright. So you wonder why is it so important that you are here taking time away from your busy time.
You know, I've been in CG for almost 25 years. I think Charles, I remember you and I were playing Counter-Strike. Remember, we used to bring our LAN boxes. Another one who used best in there. But today I know we're terribly busy. Every day, every minute, we are always working. Right? And with the admin of AI, we're now expected to do a lot more, right, with less resources and using AI as a tool.
Now what we know in today's world and in the future is that without good data, it's going to be really bad. Right? AI without solid data, it's gonna be not good. We know that. Right? And the next thing we know, since most of you are technologists, is that technical folks making calls on data on their own without the right business content?
It's also not good. Right? And in the 20 plus years, 35 plus year close, I think, I've been here in CG. It's always been us technologists making calls on data.
Right? Never a business user. For those of you who have been here, does that make sense to you? Do you agree with that? Right? Well, you know the SQL query, you should give it to me.
Well, it's like sensitive data, can I? Well, it's you know, you're asking me to create these position report, but how do I link this to the right primary keys in our company master?

What's our company master? What about our sales data? How should we link that to the positions? The client positions. Who knows the authoritative source of that?
What's the right logic? Right? If you haven't kind of done that, you figure out that you did that because of what you know. Not because of what's prescribed or what's available, correct?
Right? So right now, if I want to ask any of you, can you find me quickly within three seconds? The portfolio positions of a fund, BFA. And with strong confidence as the right table.
Can any of you do that right now? Or global client group. Okay? Give me an example. Sales flows. Can you find that real quick? Right? But if you take a step further, can you take a new person, a Trex associate, and say, can you find it within five minutes?
What do you think the answer is going to be? Check with AI. Check with AI? Well, even AI. Can you find it? Can we put out? Yeah, okay. By the way, we can't connect log data written out.
So it's policies. Okay, but anyway, so even with AI. Can it find it? And can you trust that it finds the right thing? The answer is obviously no.
But when we're done with this, it should not be known. It should not be no. We need to be in a world where we have self-service on trusted data across the firm. And we must make sure that those who are getting access to that data are allowed to see it. These aggregation policies are there. PII is protected.
Right? And both humans and AI can actually with trust and with speed get to the data leader. Right? So now, how do we get there? We want to look on the screen. When you hear the word DXP, we have distilled this operating model into four persona archetypes. And you're one of them. You're probably most the most important one.
And we'll talk about that in a second. So what are four archetypes? Consumers. That's everyone who wants data, including you. You're a consumer too.
But your business partners and everyone, business management, they want data.

Right? So that's pretty clear. What's a platform provider? A little bit of some of the folks in the room here are platform providers.
But what's a platform provider? So up until the last 18 months. What's the platform you should use to build your data on? Is it laser? Is it Caspian?
Is it EMR? Is it Databricks? Is it Dremio? Is anyone know the answer? Right? It's all the above. Right? But when you have all the above, what you lose is clarity into what you should be doing and why.
So forming DXP is saying we have a platform provider team that's Edo. Edo platform team. That is now saying there is one way to create data. The right way, in accordance with policies, standards, and tooling.
Then it makes everyone's lives easier because they're not guessing. It's not choose your own adventure. It is. This is the way to do it. Right? This is the way you do it because we have the best of the best tools.
Right? Data breaks, that's farming. Right? We are gonna make sure that people can put data in the right places in the right catalog. And we're gonna make sure that you're all popular metadata.
Why is that important? So that consumers can find it easily. AI can find these. So the paid path, what you hear, is what the responsibility for platform provider does.
Owners. So I opened the talk earlier with owners. Owners are business, senior business people. Okay. Now, for a long time, the only people making calls on data were IT.
Right? And most recently, Edom. That's a bad position for technologists. Right? How do we know here in this group if we're allowed to share certain tables on the employee directory?
How do we know if we're allowed to share certain positions of a fund to clients? How do we know that? Do you all know that? We don't know. We're not the one creating this data.
We're not the one with the policies in our minds. We see data through a table. And it's a I see table, I see data, I'm using it. Right? That position to be for

technologies, if we're making the calls, because then we are accountable.
If we take a table and we set it outside and we shouldn't, that's on us. It's unfair.
Right? So that's why we have owners. A business data owner is the one accountable for the data in capital group for their domain.
So we have a business owner for client data. We have a business owner for results for positions for attribution for trades, for research data. Right?
And they are accountable for anything that goes on with that data, whether people can use it, whether it's it's the
Right? So the vision here is we have clear business owners assigned to domains.
You can go to the DXP forward slash in your browser. And there's a link there for owners, and you can click on there's a link that shows Nicole can send this out, right?
There's a link that says here are the data owners and stewards for our domain.
Those are the business people who are accountable for the data. But now as we know, okay?
A business user with that much accountability, what's the first thing you're gonna say? Well where's my data? What's the state of it? Oh, by the way, I got a bunch of consumers asking me to give it give them this data.
I don't know where to start. But I think you're all here in this room. You're a product creator. You're a data product creator. You have the most important job in this operating model.
Because your job is the intersection of using the platforms correctly, knowing what the owners is prioritizing and understanding in terms of what is good, what you should work on.
And understanding what the consumer is looking for. And when you take all of those three things, and you distill it, you do what you're really good at, which is what?
Build these stuff. Right? We're all in this room because we've built stuff. Right?
But when we build stuff, we want it to be well informed. We want it to follow standards.

We want it to be reusable. Right? Because if there's one thing we don't like to do is build stuff and have to rebuild it again the next day, right?
We as technologies, we build stuff for scale. The difference between us and a query writer or a BI writer is what? Right? That's possible. Right.
And we also build stuff for scale and resiliency. That's why we're technologist.
Okay. So product creators is why you are here. And we want to make sure that you are well versed in the basics of how to create a data product.
Okay? And how to use the platforms provider tools. That's kind of the focus of this five days. Now, you're all going to come from different skill sets, different levels of what you know.
For some of you, you're probably yawning through the whole thing. Right? Because maybe you'll learn a couple of new things that's important. For some of you, it's like this is wow, this is overwhelming.
That's okay. So we've designed this course just for the entry and intermediate. But everyone must pass it. Okay. So by five days, you will know the basis of creating a data product.
When you create data products in your domains, and you publish it to data bricks, what's going to happen? And when you publish it with the right metadata, what's going to happen?
Okay? Now, one thing is we want to make sure that everyone gets access to that with a with the right access controls. Because you don't have a database we can table follow-level security, right?
Now you're publishing what you created to a catalog. Now that only you and your teams and other people doing the consume that data can go to. Instead of going to a person to say, give me this data, you say just go to this catalog is there.
I already voted. Right? So your job is to create reusable data products for the firm and for your site area. Okay? We're gonna give you the basics so you can do that.
Now, no amount of training will be a substitute for a real life. Okay. So when you do this training, think about your real life use cases, and after this training, put it into practice right away.

Right? Hopefully, when you get out of here, you will not be dealing with access issues. You will not be dealing with what's astronomer, you will not be doing what's a catalog.
Right, Michael? Yes. And all your questions to get started should be answered. If they are not, ask it. If our gracious instructors don't know the answer, we will log it.
My team will. Our team will log in. And if any of us need to come here and give you straight answers, because you're not getting it from the materials, Nicole will call me.
Okay. Any questions? Yeah. Alright. I'm here for questions for like five more minutes if you want. Um but otherwise I will leave you guys alone.
Yeah, lots of questions in the last cohort. They tend to save them for when they're inconvenient, don't you? You wait for a moment where we can't answer the question and the answer.
Yeah, it's okay. Uh but Martine is coming on Thursday, so Martin does a nice, it's got a nice hour, I think. Okay, of going through the entire pay path and the reasoning behind every tool choice, and that tends to mop up a lot of the questions that people have.
Good. Yeah. So that's great. Okay, thank you very much. All right, great. Thanks, everybody. Thanks for uh, good morning everyone. Yep. Where are we?
We've we put this with the matchup on. I don't need. Now, uh, did you bring you brainstormed something, did you? Yes. Yes. Yeah, thank you. Bit more monotone, bit more board.
Did we do that? We go in more board. Yes. Um, you didn't write them down though, did you? No. None of you did. Which suggests the fault is with me.
So I need to become more belligerent. More demanding. Okay. Um I've just turned off the projector, which is the exact opposite thing I want to do.
What happens? Uh, what we'll do is we'll go around and we'll get some introductions from you all. And I do want you to maybe maybe every time we talk about a data product on the course, possibly, maybe.

Write it down. And the reason will become clear, which is that I'm gonna ask you to do uh a couple more things around uh data products. And it's useful if you have them written down.
Because if you don't, then people go, well, it took us a long time just to think about what data product to talk about. And we shouldn't. Okay. Right, let's guess let's let's let's do a bit of a debrief.
Then so, ah, right, where are we? Um with me for a second whilst I wait. We're on day one. Just this on. Yep, it's on now. Okay. Cable's a little bit weak.
Cool. Um, right. Come on. So we're on day one of data creators. Call this DC.md. Day one, data creators, we're in Irvine, it's the 12th of October, it's Monday. Right, here we are. So, uh, what we'll do, remember what the questions were. The questions are what teams or roles do you have? Right? They were what shared hobbies, hobbies do you have, and then what data products do you work on?
And what were their pain points? Pain pants, pain points. Uh that you have. Yes? Remember that? Yeah, remember that. We're not in good. Um so we also need group names.
So what's our group name? We didn't talk about it. So uh so group we didn't talk about it. What teams role quickly? Teams roles, yeah. Edo. Digital.
Digital, what does digital do? CHP, I mean, like when maybe put a website. A customer facing apps and things. Good. Anything else? CXT. What does that do? Uh client experience technology. Right, client experience technology, CXT good. Uh great. Okay, uh roles. B and B Data Engineer, data product manager.
Data engineer, data product manager, interesting. Interesting. Team me. Team lead of monkeys? Technical team. Technical team. Very similar to leading a team of monkeys, right?
Um like a small capuchin monkey is like le little lemas, just getting over scurrying around everywhere, doing things on their own initiative. How dare they?
Uh what data products do you work on? Okay, let's talk about the data products. Uh client data platform. Client data platform. Tell me about the client data

platform.
Um it is a platform that CG's working on to align all of the client's data into one centralized hub. Okay, that sounds at once dangerous and useful.
Into centralized uh hub. Good. Um what's the issue with that? What's going on there? What's annoying? Um I think just the teams that own different data domains aligning together.
Um, aligning all in that, I think the business needs as we retire with one platform and then agree into another, I think, you know, just trying to figure out like misunderstandings or discrepancies across the data.
Is it also gatekeeping there and uh loss of ownership sense of that, maybe? I I think the teams that we work with have good ownership. I think it's just aligning. Just alignment. Okay, okay. Good. Because when I when I hear about like, oh, let's have a centralized hub, different teams are in different domains.
I hear how are being taken away from different teams. But you're telling me that isn't an issue. Yeah, it's like one team has like retirement plan accounts, another team has investor account data, and it's just how do we bring it all together?
Cool. Okay, great. Um next data product, any more? Positions and transactions. Positions and transactions. Um tell me about that. We are trying to build a A-bar positions table from accounting click bar.
Accounting book of record. Yeah, yeah. And then finding sourcing the data, figuring out what is the right for each kind of product that we own. Okay, so sourcing the data main pain point.
Okay, cool. Any others? No. So we interact like uh we captured what uh users are doing on the website. So the pain point, but we have like we have a raw data, but what Edo is doing from massaging on, we don't have any like red long inside it. So what we got is like the curated data, right? But sometimes we want some data, but they have our bandwidth issues and whatever to expect. So getting getting work from Edo.
Yeah, we go up with the EDOs closely and then uh get the raw data massage in the form which we can use in our CDP. But you presumably you have the raw

data.
So you're looking for curated reports, is that right? Uh curated raw data in the form which we can use in other played forms like CDB, like the tested target. Yeah, yeah, so so if you okay, okay, okay, okay. I'm gonna say getting data from ETA. Uh before we deep level now. Okay, great. Uh we'll come. Uh maybe we'll just do the couple from the rest.
I'm not gonna uh encourage everyone with the same level of depth, otherwise behavior. Uh back of the room there, what's your what's your full name?
Hi. Hi. Okay, that'll do. Uh and uh okay, so teams roles. I am team. I have investment operations. Okay, because uh investment operations? Yeah, okay.
I mean, I know another hero asks for some teams and things, so we've kind of done this, but what roles and what roles? Software engineer, okay. What do you what's software engineering is there in investment operations?
I'm working on some report or subject or IMM. Right, okay. So those are capable system basically. Right, okay, part of the topical investment system, sure.
So they could cut uh like uh B2C app or B2B. Oh, internal application. Okay, that's fine. Um other roles in the group? Edo. Edo what do you do for Edo?
Data operations. Data operations. Cool. Um digital marketing. Digital marketing, and what's your role? Solutions engineering. Solutions engineer, okay, interesting.
Uh what do you do the solution engineer do? Um then from DC own um basically marketing, so you need marketing locations. Uh huh. That go out of capital group.
Right. Um platforms. So yeah. Cool. That's just an example. Okay, great. Any others? Roles. Two move on? Okay. Okay. Data products. What data products do you have?
For the marketing communications, um, that does the engagement data. Okay. Email engagement data. Uh marketing data. Any other uh what are the pain points there?
Um so the platforms basically just um data from different bases. And one person

can have multiple like profiles. Uh-huh. So I see. I believe just reconciliation of that within the uh the downstream systems for reporting.
Identifying one customer or one person across multiple events or um data well as records or anything so that. Yeah. So um in policing that's called a golden nominal.
And in policing it's very, you might have heard the phrase golden nominal. But um uh means it means uh unique name, basically what it means. Um and in policing it's very hard, of course, because you have frontline policing staff just writing on bits of paper, so they could misspell something, they've got different different names of things, you know, Irvine with you know with a Y or something, who knows, right?
So actually, even understanding what um unique thing in the world that they're talking about, it's very difficult. So similar, similar uh issue here.
Okay, cool. Um any other data products one more? Yes? Um uh NFR search. Enterprise search. Digit uh so we have the digital public reason digital websites. Yep. And um we have a search engine on there that we collect a lot of um search for the related data. Um like to right now it's pretty i but um it's pretty siloed and then the vendor's platform.
So we don't have a way to right now we're not currently we say vendors platform is a vendor providing a CG website? No, no, it's just a search. Oh the the vendor's providing the search.
Yeah. So we have this all this search data that's kind of like. Okay, you agree. I like that. Um should we move over to the the table of uh extreme agreement? Should we call you back? The group the table of extreme agreement. There we go. Uh so uh teams roles, hobbies, maybe roles, we've looked at teams a little bit. I don't know, I don't really mind. But um just something to introduce yourselves a little bit. Uh we haven't done hobbies really, that's fine, we'll skip it.
Um then data products. Hello. Roles. A few of us were from EO and mobile technology groups. Okay. And what roles do you have? Data engineer, yeah? Data product manager. Okay. Good. Product manager. Product manager, great.

Uh data products. Data products. I support attribution. Attribution? Yes.
So tell me about attribution. Well. What is the data product there? What's what's the underlying data product? So it is um the equity, multi-asset.
Basically, it's data attribution. Like um we we usually have like a monthly and quarterly uh that's being published from the source system. And to downstream. And so I'm supporting it from the task family. Yeah. And uh what's some pain points with that? Currently we are um working with a business to kind of um take a look at all our data sets and removing you know some of the duplications on the different data set, trying to align it for the target state, which is PayPal.
Okay, good. Um who else in that team? The product? Sure, so all the product reference data. As well as the capital multi-portfolio system. Yep. And indexes. All the reference data master, morning star data, yeah, investment data. Just all the data. All of it. All the data all the time. Um what's some pain points with that? That it's all the data all the time all the time. Everything everywhere all at once. Yeah. And the social networking that has to happen to get to all the data owners to align.
Yeah. And across what Hero was saying about all the different lineages and versions. Yeah. That have to be supported in production at once. Mm-hmm.
Um people make some people may know. There's a famous problem in in microeconomics, which is why do business exists at all? Why and why not just have independent contractors and just sell your labor on open market, right? Uh, one of the answers to that question is because it reduces transaction costs. In other words, there's a certain cost of the certain cost just to turn it up to an office, right?
And you have to go to a cafe, right? Especially in LA, right? Might want the what cost there is just to get anything done, right? You have to be in the same place. It's very expensive. Um so businesses exist so that you can reduce the alignment. You can think of it as reducing alignment cost, right? Which is how do we get the alignment all the way down?
But a certain size of business, then you start how you start essentially having

those kind of huge transaction costs again, right? So if you're in a small bit like a hundred people business or 200 people business, you're essentially getting free alignment.
Right? But once you get into thousands of people, you then you send you then almost getting back the same cost you'd have if everyone's an independent independent contractor.
It's like, oh, I need to call the independent contractor in the New York office or something, right? And then I'll get it get get ours and get a zoom call, everything else, it's you get back to the same kind of costs.
Um which is then why you get bureaucracy and why you get you know you start it's like having a bureaucratic system to try and push the transaction cost down again.
Um but yes, so alignment becomes one of the biggest issues as businesses scale, right? Because then you're very expensive to do any. Cool. Alright, good. Um we're kind of overseed the customer data platform. Customer data platform? Data management platform. So most of our work is trying to sourcing different data from our across the organization.
And yeah, you're kind of activate all those. Okay, cool. Um one more group, is that right? Not going crazy on that. Oh yeah. What's your group name?
Oh, group three. Um right. Roles, teams, products. You know. You know? What you do is what do you do? Yeah, what do you what do you? Um need a DPM? DPM? Okay, what else? Um we have software engineer and software engineer, and we have an architect. Oh, what kind of architect? What kind of architect? Technical architect. I gotta get paid more for that one. Easy architect. Ah, yeah, I was gonna say, what's the architect? Data architect. Ah, right, okay.
So because my as you may have heard in intro, my background is data science soft engineering, principally soft engineering and data science wrong.
Um so when people say architect, I'm thinking technology. Oh, solution architecture. Uh I mean, obviously in this context. So I know the load of solution architects.

Oh, and the solution action. Okay. Because like I mean, you know. Yeah. Oh, and uh uh, yeah. I I I I would find a technical and solution I can. That's my point.
Uh uh, okay. So um anyone else? Any other rule roles? No. DBM, SM. DE. Uh solution engineer as well. Solution engineer. Cool. So part of the reason I'm doing this is people so people can see across the room that we have DPMs in the room, we have a mixture of profiles as these.
So people are asking different questions, or if I'm saying things that sound like, well, why is he saying that? It might be for a different audience, right?
Okay. Um data products that you've been working on or know about. Ums product. Assets and you say flows. Yeah. Flows of clients. Okay. Walk me through that.
What do you mean by that? Um the um basically what are the client sales for the engines and and all the sort of things. Right, so so uh like like by tracing uh client action through the lifecycle, like a client license.
Yeah, yeah, trans action. Uh okay, any other data products? Client profile data. Client profile data? Might sound similar to client data, but yeah, we mastered source client profiles.
We must student source client profiles. Master and source client profiles. So like uh candidate client profiles, like or reciprocate rules. It's the CGI's clients.
Right. Okay, so is that retrospectively? So if we we all have these kinds we have these clients you want to know what profiles they match on. Okay.
Um we do a lot of deduplication, data quality, okay. Okay, and then some pain points. Any others in that group before we do some pain points? Okay.
Some pain points then. What's annoying and frustrating about the things that you work with? Data quality. Data quality, what do you mean by that?
It can be a very specific, almost too specific example of the data quality issue. So many times we don't know whether it is the right data or not.
Is that data quality issue? Yeah, from external sources. Do you mean you don't know if it's fit for your product, or you don't mean that it's a correct?
No, whether it is correct or not. Correct not right, yeah. Okay. Um other data pain

points? Why is that? So what causes that? What causes the lack of correctness? Is it just that small you have small vendors and they you're waiting on an Excel file, or what's the um it's not just that. Um sometimes it's not one product which can tell you uh you have to look into other data too.
Right. Correlation uh to understand whether it is right or not. Right. So um individual products are not a one, and the correlation is where we can read all whether we have done that.
So that correlation doesn't take slightly. Cool. So but what in what sense could it be wrong? Um for example, uh, I'm gonna give you an example.
Yes, please. Um you do 50 billion Ts. Yep. You don't know whether it is right or wrong unless you see the uh UN. Okay, okay, okay. So it could be line to you, is that right?
I mean, we know it is 50 billion CNC wrong because one day no one can see it. Right, so that could be it could be just uh uh 50 million rather than 50 billion, but it could be an extra zero.
Or you could okay, so it's a data entry issue in the end, but then but okay. Um different sources are issue, right? So someone might be sending an email every now and again, and you're waiting on the email, and then you don't know if it's coming from the right part.
I mean lots of different sorts. I'm just curious as to as to an example. Okay. Um right, okay. Um what we'll do is we'll we'll we'll discuss some stuff now.
Uh I'll maybe I'll introduce the course actually, and then we'll discuss some stuff, and then we'll take a break. Cool. You mean you've been listening for a while, I appreciate.
Um so yes. Um what is this course? So the on Friday, we have a capstone project, which is where you go away and um you know, create a data product based on some CSV file data based on funds and all that sort of stuff.
And the goal of that product product project is for you to get a data product onto uh Caspian uh data bricks and be certified that you we've observed you to do that, right?

So there's a phase one of that project, which is a kind of certification bar barrier if you like. If we all if I observe you to do that, then we've ticked off that you are a data creator, right?
Um so the point of these four days, which are Monday through Thursday, is to prepare you to do that project. Cool. Um so what do we cover? So today is a kind of soft skills, mindset behavioral day, group challenges, things of that kind.
Part of the reason for that is to cover off or to ask the question, how do you design a data product? How do you design a data product? And that comes down to you know, you know, interviewing people who need the data product interviewing customers and uh thinking through how how products get adopted and all that sort of stuff, how it gets finance, what business case would be.
So you might think that's primarily for data product managers or something, but it's not necessarily because technical stakeholders play a vital part in that design process, right?
Then on Tuesday, tomorrow, um we might have a couple of things in the morning to continue from that, but then we start introducing technology stack pay pack in a lot of detail.
Um Docker containers, might have a look at Git there as well. And we'll look at the entire paid package system. And then on Wednesday, we're focusing on one core tool called DBT, which is how you uh in some sense how you literally create the data products on the page path.
It's very SQL-like. And then on Thursday we focus on Airflow, which is how you orchestrate how you how you sort of queue up and schedule and then do the SLA stuff if you like and print those terms.
Once on Thursday, and then we go into Friday, which is Cash Done project. Cool. Um now there are other technologies and things we'll talk about data bricks, we'll talk about uh we might even talk about like harness and some of the deployment context around um data products.
Um those will just be slot in around our other conversations. But I've given you the focus. So the focus of data uh Monday is how to design the data product,

focus of Tuesday is kind of what's the paid path, how do I do things on the page path?
Wednesday is how do I use DPQ? Thursday is how do we use F1? Friday is let me go and create a data. Cool. Cool. Let's ask one question and let's ask and answer one question, and then we can go uh for a break.
So uh the question I want to ask before we go for break is what is a data product? Shall I should we see what happens if I use a whiteboard marker and uh what is a data product?
My writing is not an excuse. Okay, so there's a series of properties which we can call promises associated with the data product. And uh we would say that it's trusted, reusable.
Business ready data sector. Business ready, uh there's some other things there as well, which is that it's accessible, right? Accessible, trusted, reusable.
Okay. So what do we talk about? But these are properties of what? What are they properties of? Like not property of car, right? Car should be trusted, reusable.
Right, but we don't mean car, what do we mean? Of what? What's being trusted?
Data. Data, yeah, so some data sets, yeah. The promise is, so you can think of it as a data set plus a set of promises, right?
And I can show you the ones at CG, that are researching the illicit update basically, we use more shareware. Um good. Um promises are enforced.
Contracts.
Contracts, okay, good. So there's written contracts, I suppose, and design documentation. How else are they enforced? Data quality checks. Data quality checks, so with code by coding for jobs.
So we might also have a kind of uh so some of these promises are enforced by code, which runs on a certain infrastructure. So one view is that data product is a data set plus set promises.
Another view is that a data product is some data plus the code infrastructure that enforces these promises. So from uh from a platforms or engineering point of view, that might be the when we say data product, it's what we mean this.

From a data product creator or product manager point of view, we're thinking of these terms, make sense. Okay, cool. Um there is another view of what a data product is.
Is it any data set? If I take uh morning star data set and I just sort of drop it into a folder somewhere, even if I even if I open piece of drop, right?
Is that a data product from a CG point of view? It's a data product from a morning style point of view, right? That's probably a data product. But from a CG point view, that's just a source piece of data, right?
So there's some that's not really an internal product, right? That makes sense? So there's something missing here. What do we have to do to the data to make a day product?
Actionable. Actionable? Okay, good. It could be. I'm thinking in terms of alignment. So that again? Business purpose, yeah. So I mean, but I mean that's registered alignment, right?
So one thing I do, another view of a data product, is that it's a transformation of a data set. Not just a not just a data set, it's a transformation, right?
And the transformation is that we're going from a source aligned data set to a consumer aligned data set. Make sense? Okay, cool. Now what does the word alignment mean here?
And then we can take a break. Rush action. What is the word alignment mean? I guess it means the goal, sort of thing or uh. Good, good. So I like that, yeah.
So there's something to do with goals there. And importantly, there's something to do with people. You said team, you said person, etc. It's not a technical, it's not a technical condition.
There's no such thing as a source aligned or consumer aligned data set as a technique. There's no code you can run or test you can do on a data set that tells you the source aligned or consumer aligned.
It's relative to a personal key, right? So what source aligned means is that there's some people over there, and because they're over there and they're not us, they're evil, right?

They're the other people, right? Um so they've got they've got their own interests that we don't care about, we don't care about those people. But they've done someone in terrible to us.
They've given us a data set aligned to their interests. How dare they? So then we have to uh we have to spend time and energy turning up to work and typing on a keyboard to fix their problem and make it aligned to our interests, right?
Or at least our customers' interest or consumers' interest, right? So um what that means is that we have to, of course, apply a series of transformations in roots of data, apply rules, or the whole design piece and the whole uh notation piece.
So what is the code doing? You could think of the code as enforcing the promises, but the core code also does the realignment of it as well, right?
So you've got code that does it if you like SLAs or promises, SLAs, and it also does kind of realignment. So these are kind of three views of what a day product is.
One view of data product is a kind of just, well, you get all these nice things. That doesn't really tell you why it's valuable necessarily, it just gives you the promises, right?
The value comes from the realignment, right? The fact that you've realigned data such as you interest. And from an engineering point of view, you might just be very confused by that.
Okay, so we've realigned some data and it has all these promises, but I'm an engineer, I don't know what that even even means, right? So from an engineering point of view, it means that you have this code and you have this data and you have this infrastructure, and what the code is doing is doing the realignment and it's doing the SLAs and the promises if you like.
And what the infrastructure is doing running code and prediction data. That makes sense. That's three separate views depending on who you are. Cool.
Now, what a data product isn't, a data product isn't then just a pair data set, right? It doesn't have these things. Um data product isn't isn't just some piece of code, it's some infrastructure, right?

It has to be doing these things. And what a data product isn't, isn't just a source data set. And it isn't even just a consumer aligned data set just stranded out there in the ether.
It has to have to be a realignment. Cool. And so in the provision of data products, what we want to do as a product creators is provide those things.
Provide coding structure and data, provide a realignment, provide that. Right. Tell me we all. Should we could should we back come back here at 10 30?
10 3 0. Cool. Alright, I'll see you back here at 10 30.
Alright, good morning everyone again. So we're gonna answer the question now, I suppose. Uh, how do you design a data product? I can zoom in if you like.
Um so what questions do we think are relevant when the initial phrase the very outermost initial phase of data product design. In other words, someone comes to you and say we need a data product, what what questions are you going to ask?
I guess what's the scope? Like what areas are covered? Yeah. Uh sure. Can you put scope down there for a second? Phrase has questions, but what's the scope is a little bit too broad?
That's good. Uh what else? What else are we asking? Source of business problem. Pardon? Business problem. Business problem, yeah. So what what's it for?
What's it for? Yeah, keep going. Source of data. Yeah. Sure. Who had it done do you want? Yeah, I mean, uh I mean this is the this is I would say this is the next conversation rather than the very first.
This is getting you to the design of the product. So I'm thinking the very first conversation, someone says you need a product. What questions are we asking? Yeah, so who needs it, I suppose, right? Who needs it? Do we already have it? Yeah, okay, good. A business purpose. Do we already does it already exist? Does it exist? And relatedly, what are people doing now? Even if it doesn't quite exist, there might be some work around people are engaging. Yeah, what else? Keep going. Why do we need good? Yeah. Do we still need it? Do we still need it?

Yeah. Um, okay. So these are questioning what to some degree, what it is and what we're trying to build.
Suppose we answer all these questions. So what now? What are we what questions are we asking now? What tools do we need? What tool, yeah. Um, so I'm gonna say can we build it?
Can we build it? And then maybe how do we build it? And then how do we build it? When do we need it by? Yeah, when do we need it? Oh, that Japanese quite. Do we already have the data for it? Yeah, uh um does the data exist? So these are all related to can we build it? How do we build it? So that that would include what's a data source and things like that, for example.
How do you build what a data source? What's the transformations? Uh what's the platform? Sensitivity, PIR, etc. etc. etc. etc. Yeah. Um is it to be delivered?
How will it be adopted? Why will people use it versus existing actors? How frequently does it get updated? Yeah. So I'm just trying to think about where that goes.
Um grain of the data. That would be how how. Um suppose we answer all this. That's still um now we're at remember at the outermost conversation before anything has happened here, right?
So we need to make a decision. Are we gonna go ahead and build it or not? Right? I'm gonna go ahead and build it or not. What questions would come into that what into that decision?
What question comes money? Good, yeah. So give me this needs to be broken down into something, right? So I'm gonna break it. Is it worth it? Is it worth it? Is it viable? Uh is it worth the cost? Um is does it deserve, if you like, does it need this particular level or prioritization? So, i.e. build it now, right?
And then is it valuable? So you got you've got desirable. And this a lovely scale question. Right. Are we just doing it tactically to get something done right away, or is it gonna be here for a while?
Yeah, what's life have value?
Yeah. Um couple of things you can do now. A couple of things you can do now.

You do a group challenge around this, answering these questions. We can we could look at interviewing people.
Should we do I let's do a group challenge? Group challenge. Let's do a group challenge.
Okay, so um, so here's the group challenge. The group challenge is pick a data product you've discussed this morning. Or if you're feeling ambitious, a new data product you propose for CG.
And answer these questions. However, I want you to focus on the non uh the non-technical areas. Um primarily. So who needs it? What's a business purpose? Does it exist? Can we build it? How do we build it? I want you to prioritize that less. Cool. And the reason is that we'll discuss how you can build it for the rest of the course.
So I want you to think about how you would justify the creation of debt product. Cool.
Alright. I'm gonna give you a little while to this. Uh so it's 10 40 now. Script challenge, what do we want? Do you want to half an hour? Maybe twenty minutes.
Should we do twenty minutes? Eleven. So if you get done with one data product, you can do something incredibly ambitious and choose a second data product to talk about, right?
So uh and when we come back, I want well I want people to uh when we debrief together, I want people to reflect on what was the most interesting, surprising or difficult question to answer from this set.
Cool. Again. No group, group group. I recommend use the paper as well. Like assign a note taker, take some notes.
Oh, it's not my own. I don't think that's the definition.
Okay. So name. Table three or something. Cool. So rather than um go through your whole conversation, two questions. First question is what was the data product you were talking about?
Second question is just across this space of questions, which uh part of the

conversation was difficult, surprising, or interesting. Cool. Yeah.
So what was the data product you were talking about? We were talking about advisor data. Advisor data, yeah. We're good there. We'll find out when you continue speaking.
Alright. I think um to summarize it, one of the most difficult, and then you guys speak up as well. The time that we spent a lot on is probably um the adoption.
I think you kind of circled back to all of our other questions regarding who needs it, does it exist? I think that kind of all wrapped around adoption and how is how do we adopt it for other teams to consume it in scalable way?
Cool. So tell me, okay, so we've got advisor data. Remember that data itself is not a debt product. So what are we doing? What's the business purpose of this? What's the data? CEM. Right? Right. See, this is C go ahead. There you go. So it's to maintain the client profile data corresponding to our clients who are advisors. And distributed across the enterprise. And what how is this data? Is this data being provided to someone in a dashboard or something? Or uh how through it's through APIs, through files, okay, group Aspen.
We already do that. Yeah. Um at scale, but there are problems. Sure. So you've got this AP, so from your point of view, the the API is like your consumer rather. But um, how does that where eventually does that go to? That must go somewhere and user. A lot of enterprise controls in that API. So it's contact, let's say CG's contact center, digital websites, right?
Yeah, okay. In the digital portfolio, then CRM marketing and sales. Okay. I see. I see. And what decisions are they making with that data? What's the very the end of the final end user?
What decision are they making that thing? So sales people use this data to rely on the data to make better relationships. Better plan relationships that helps leadership with uh when they look at analytical dashboards, leadership gets a view of um, you know, whom are we interacting with the most, how are we building better business with some of the larger firms?
So this is essentially uh and this this product here is essentially an internal

product. So you have to get other teams to consume it. Right. That's the adoption thing.
Um what are the barriers to adoption then? What are the issues with adoption? Um it's mostly how the these consumers view this data set. Some of them look at it from a I mean there is diversity in how this data is consumed and viewed. Some of them look at it purely from a marketing lens. So they all that they're interested in is name, phone number, email, that type of stuff. There are others who are more interested in other aspects of it.
So the dimensions vary quite a bit. Yeah, can you be some under the lens? Another lens is mostly what are their affiliations, right? Could be trading identities, who are the most what are the most commonly um surface products?
You know, these advisors. Okay. Um, so you've got different competing interests for data and that creates adoption issues. How does that create adoption issues? It's mostly um the APA contracts and data contracts, they're a great project. I think at least what we expose. There could be more flexibility. They we are expecting to be more flexible, but yeah, the the contacts are rigid, so different consumers we we do a lot of juggling to satisfy the client means.
Cool. Alright, thank you. Alright. Uh what's your group name? Eh? We didn't do it. We didn't talk about it. We didn't talk about it. Now that was almost a question there, it was rising in the end, right?
Yeah. Uh okay. Um what was the data product and what was the most difficult? Surprising, interesting question. Uh we chose the same thing. Does you guys show Millie?
Of course. Okay. Uh CDM. It's called CDM. Yeah, tell me about it then. What's what's that? Um distribution of client relationship. This means how capital sells the services, the financial advisor.
Offices and forms that sell it funds. Uh the table field, like the office type, RIC update and so on. These are the things which are I need the volume levels change.
Okay. So my my uh my right here is not getting enough, my left ear is getting too

much. Okay. Should I repeat? Yes. So the business area, the distribution and the client relationship.
Yeah. This means how capital group sells to the services. Yep. The financial advisor of the and the firm that sells it funds. Firm that sells it funds.
Right. The the business problem we are trying to solve is one trusted current record for every advisor office. One trusted record for every advisor office.
We deal with who they are, whether they are active, and how we can cover that.
Okay, great. Okay. I understand that. And yeah, question. So you have the biggest challenging question.
So who needs it? Who owns it, right? So the consumer would be sales, wholesaler, internal sales, death, uh the client services, need the right officer status.
Marketing, uh compliance, uh sales finance. So everyone needs it. Yes. Now the ownership. Uh again, it's like the business owner, the head of distribution operation, data short and technical owners.
Like the Edo data product. Yeah, I mean that's me. That's not good, that's not great. I mean, is there is there needs to be an owner? I mean if you know one mean though, I don't know how to share the PA information, but yeah, share in the it's in the UXP forward slash thing.
I think it's it. Okay. But well. I'm sorry. You're feeding the Synto A model and then trade it with all this information. Oh, you're doing that.
Okay. So do you have a little copy to this? Yeah, give me a proper copy. There's a transcript taking place. Is it transcript? In fact, when I give you the notes on Friday, I guess, or some of the day.
Uh the uh you'll find the notes even have heroes in the beginning of them, which they currently don't. So I'll I'm uh uh I'll put the transcript into the notes so that they're they're in the notes.
Um okay. What else? That'll do, right? Okay, so mix yeah, so uh need, okay, great, let's get keep out keep going. So uh middle group? Disagreeables.
We're disagreeable to negative. I disagree. Okay, then we'll agree that we'll agree

that we disagree. We'll agree that we'll agree that we disagree.
Okay. I don't like that. So we chose the same data set. Yeah, that's fine. Um so what's different than everybody else? Yep. Um we talked about globalization as an issue.
Okay. Um what else did we talk about? That's what like fragility in supply lines across integration between Canadian and American oil. Yeah.
Okay, tell me about the tell me about globalization in this context then. Actually, tell me about it. In that. What's the issue? What is the issue?
Well, the issue that historically we're very North America centric, and now we're trying to globalize, and those other operations are less mature, slash smaller, nibbler.
Yep. Then maybe a hidden user group is the uh global product development group? Because they're actually looking at these customers for market sizing and new product development.
To say what can the market bear, what new products should we offer these people strategically? Yada yada. Cool. So what specific issue does a less mature other office create for you?
What are the issues that creates? So that's data standards, data, so lack of maturity data standards. Okay. Structure. So it's just that translation of love for things.
Okay. So it's just like when multiple hospitals problem or something that just everyone's working some different. Uh but I think that well, I think the other one, the really big one that we've got also is regulatory.
Yes. Okay. In that space? Fair enough. Regulatory compliance. You would think that uh in a um you know universal healthcare system, all the hospitals would be working working from the same data model, wouldn't you?
It's just the same, it's just the same as any other system. Uh okay. I guess that's a problem with um kind of bringing uh you know early comp early computerizing of uh systems, you know, early 70s and 70s stuff, and then you get everyone doing anyway, good.

Um the issue there's globalization, what kind of issue is that? Is that like um is that an adoption issue? Is that is that what what kind of issue?
They're just like a getting started issue, how do we build a kind of issue? The centralization and scaling all happened in one region, one locale, and now all of a sudden we're trying to get everybody else in and they don't follow the same model or rubric.
Right. So it's a kind it's a mix, it's it's an issue that clo cuts across several different concerns. One is can we build it because we don't we don't know? The other is is is it feasible because if the if the value trade-off is different because you've got all these different alignment issues? Okay, cool.
Um what's your group name? Hi. What? Hi. Group no. Okay. We should say dot dot dot, dot dot dot, blah da, dot. Hi. Uh you know the little video games where there's just silence and they can give you three dots.
Like a beat in a script, beat. Um okay, so what was the day product? What was the most difficult, interesting question, et cetera? Well, you call it user experience and slash engagement.
Mm-hmm. Tell me about that. Um the uh uh users of those marketing. Okay. Uh and the owners are probably I and A. So we think. Cool. Uh it's got uh personalized uh marketing data.
Okay. And uh it's used to measure engagements? Sure. Uh with what? What you what you're measuring engagement of? E-mails, search, right? Email. Any uh any communication channels?
Cool. Right. Um what was your biggest question area? Uh the the challenge is integration of the data. Right. The data sources are scattered. You guys data is there, but um putting all that together and uh packaging it into a consumable uh product is the current challenge.
Yeah, I mean so that uh so can be like globalization. Globalization, yeah. So what's the problem about that uh this is the problem with building it, right?
I'm I'm not that much of an I understand this problem with building it. Um how does this translate to a design question? Do you have a question I think?

So it becomes what becomes less more costly, becomes less feasible, more difficult to design. It becomes difficult and more difficult to adopt it.
I think design is the difficult part. Yeah. For example, all the data that is relevant to experience and engagement right now sent in silos, right?
We have the email communications team sends to laser its own way. Research has challenges of getting into a capital group ecosystem. Um maybe we are not sure.
So the research of how each team individually is sending the data and redesign is what was the most challenging discussion. But we see the value uh much higher to be set, but end of the day, whatever we do has to have a clear path of measurement and a product which can be integrated right now.
I'm pretty sure all this is might be 50% there with a team, but not as like I sitting here, I don't have any idea like for what we said. So that was really I guess um what I'm trying to get at there is we can we can we can kind of that that the the the these questions fall within umbrellas of questions if you like.
And umbrella's questions are the stating one type design phase, which is if I'm trying to assess whether we build this data product, I need to understand, for example, the value type product or how it'll be adopted as to whether it's built or whether it's feasible.
So if people come to me and say I've got all these challenges, I'm going, okay, I understand that all these challenges translate that for me into whether this means we should build it or not, right?
So how do we do that? How do we do that? Um so we've got the the total value of the total lifetime value of the product. You want to have a new one talk about the total lifetime value of a product?
How do people answer this question? Well, we just said long term. Long term, that's fine, yeah, but did you give me a bit what's the jollers? Any way of estimating that?
How do we estimate that? As long as we exist. As long as we exist. Yeah, but people who know calculus will know that you can take limits, and once you take

limit to infinity, that means it can still be a bad number.
In other words, if every day from now until the end of time, we cost half as much or cost twice as much as the day before, then the value is infinite.
Cost half as much value is zero, right? Anyway, um the point is uh how do we estimate lifetime value of a product? Anyone have to type out that question?
I was thinking about the lifetime of the tools or infrastructure that we need for you. So there's maintenance costs. So that's total cost of ownership, so on the negative side we've got the maintenance costs, um, which means you have to have teams.
Um also opportunity costs for those teams, right? So if the teams are on this product, they're not on some other product. If they're maintaining things and not creating things, right?
You have to factor that in. Um expect capacity there is to maintain a new data product? Cool. What other negative costs are there to create a data product? Licensing costs? Licensing for acquiring data? Time for people to develop product. Prioritization costs, which are opportunity costs. What else?
Alignment costs, right? So you have to have SME time or the data expert time, whatever they think you might need. Cool. Any other costs we can think of?
Cost of adoption, exactly. So right now, what you're doing right now is part of the cost of adoption of the paved path, right? So we call the paid path, what's the cost of that?
Well, quite large, because you're here, right? So adoption strategies are part of the cost. And if you don't, if you don't have it, we'll talk about adoption strategies in a second.
That's my second question. You like questions as well. Uh talk about those in seconds. But yes. Uh what else? Any else, any anyone to think of any others?
Well, alignment costs. Now what are those? So like, well, you know, if we had um if I run this business, if I run a very large business, like 10, 10,000 people or something, I might even, because I'm insane.
Have like an auction market for people's time, right? So you know like some kind

of like 1970s Chilean experiment where you do have crazy capitalist experiments, right?
So anyone knows about like you know, crazy 70s capitalism. You had lots of uh people from Chicago go down to various American countries and just start doing random random things.
But uh one of the random one of the random things that I might try and do is have an auction side of people's time, and then you'd be able to price that exactly, right?
So I want to call this SME in this department. Well, they're there's their entire backlog, and you have some internal tokens in the business, and you you you ever get to allocated number of tokens for their own theme or something, or like a hundred tokens, and then you have to do it on their time.
You have a half an hour of their time. Then you can price that high, and you would know how much that costs. And then someone who knew their salary could go, oh, that's how much time that costs us.
It costs us a thousand dollars or five hundred dollars, or like half an hour. Cool?
Now, if we did that, that'd be very clear, right? That's too what things cost.
Um there's a line of cost, getting people the same room, getting people on the same call. Yeah, okay. What else? Okay, so on the positive side, uh, what are the payoffs from a uh from some of the for some of the data products you've been working on, what are the payoffs?
What do we want? Consistency, yes, it's very hard to price that, isn't it? So uh one way you could do that. Now, if this was a data science course, or we're doing statistics or something, which some degree I might prefer.
Uh we could go and we can have a little sidebar about how we might model that, right? But what we could say is it's part of the business case. We could say, well, look, um, here are all the alignment costs that exist at the moment because of our current way of working.
In other words, there's this number of meetings a week, or there's this number of types of people we have to contact, there's one change over here, right?

All these downstream consumers, one change over here means we need all these meetings to line up, etc. If we have this product in place, those those those half or those get reduced.
So we can we can price the value consistency by impact on alignment costs, yeah. Uh what other payoffs are there from creating a data product? Time consumer save, not all the data is a lot of things for.
Time the consumer saves, yeah. Again, very also very hard to price. But it's really lovely if our consumer is bringing money into business, because any time that we can allocate that for them, we can directly measure like you know, if they're making a million year or how many yeah, right?
Uh we could say, well, an hour of airtime is hugely valuable, so we we can we win. So oftentimes you may realize this is that things get prioritized in the business the closer they are to the value generating edges, because not necessarily because um the the products themselves are the better products to develop, but because it's easier to perceive the value.
In fact, the entire finance the entire financial industry exists almost precisely because of that phenomenon, which is that you get to scoop up you know you guys get to scoop up financial value from the system because you're at exactly the points of the system in which you can observe that value.
So you're able to extract some of that value. Make sense? Now if if if you were just so if you're just like sold as like musicians or like plumbers or something, right?
Where you couldn't see the value, but you did the exactly the same work, you'd make less money, right? So the the work itself is not necessarily where the value is coming, they're coming from the working adjacent to where you can observe the streams of money in the system.
Anyway, so you want to know how financial people make why do people make money from like Excel files? The answer is because they're very close to the money, right?
So we can see, right? Now what that means is that when we're going to have

business cases where we're very far away from the money, it becomes harder to justify this, right?
So we have to get creative in our identifications, like for example, you know, consistency alignment. What are the payoffs out there? What are the payoffs out there?
So you've got consistency, you've got alignment payoffs, we've got we could we help the consumer down the downstream consumer, save time. Uh what else? Can you tell me? I mean I can come up with some generic ones, but new something. You already kind of what what uh what value do you think you have? It can be a new business opportunity. New business opportunities, yeah. So there's also something about like there's also something here about what happens when good things are in the right place?
So it's not necessarily that right now, so let's say we create a data product where all the, you know, let's I don't know, uh all the you know, you have a giant product catalog where all the products are there, and we can everything in one place and you can see all these things.
It might be like, well, you know, our you know the consumer here already has that cross across two or three different dashboards, it's not gonna save them a lot of time, but it might then create opportunities in the future because things are out next to each other, uh, for those people to make you know greater inside of deep organizations.
It's hard to hard to price those things. But you can you can say, look, it's part of the job of if you're like visionaries or evangelists or solution architects, is just to create things in the right place at the right time so that good things happen in the future.
Um other payoffs? Pardon? Trust. Yeah, so reducing custom I I hear that as reducing customer churn. So what? Friction. Friction. Yeah, friction and then also operationally that you can operationalize something and scale it as opposed to inefficiently run the run the business.
Good. Bring down general transaction costs across the business, yeah. I mean

that's what the paved path is, right? So you think about the paved path, which is we'll talk about it tomorrow in detail.
The paid path is something like all the technologies that comprise this course. Like what's the point of doing that? Well, it's a bring down the transaction costs across the business everyone's aligned in the same technologies, right?
Very hard to price that, isn't it? Like how to justify that. So at some point, some people of leadership need to make a kind of visionary judgment and say, well, I kind of take this business case, and I have enough experience in the business to see that there's tremendous value there, even though we can't price it.
So part of the job of leadership is to make those kind of calls. And generally speaking, there's a good often a good kind of wall between people who can price things in like leadership, and oftentimes you find surprisingly, leadership can make better decisions than people's prices because people's pricing is very limited to the things that can actually measure, right?
So the value of leadership if you like. Why don't we just do like instrumentalize everything and just like do it based on numbers? Because well, if you try and do it, you'll find out.
Okay, we any anything else? Anything we want to say? So what would what the question we're answering now is how do I provide a business? How do I contribute to a business case to justify the existence of creation of data product, right?
And we think about the liabilities and all the all the payoffs. Costs. Anything else we might want to add to this stage? So can we think about each individual clause if we have the line single source of the code?
Good. I like that. I like that. So in the modern environment, of course, they've got like process efficiency, you could say. So we improve process efficiency by making uh data products available to AI workflows and something.
That's great. Yeah, what I like is focus. And there's not necessarily a wrong answer, but there's so many right answers, there's never a good one.
Yep. So the focus to just say this is how we're going to do it. Let's go. As

opposed to, well, that's all the way. That's the value of all the other things.
And sometimes just being on the same page and go in one way is really the most valuable. Correct. I agree. I agree with that. Significantly. We could say a couple of things about that.
Yeah, people just making decisions. Good. Good. Just trying to just sort of cutting things in my mind about things you can say. Let's move on. I might say something about that later.
So let's talk about now the other question which is relevant to this phase of design, which is adoption. Cool. Adoption. So what causes data products to fail? Well, one thing that causes the I think I would say one of the main things that causes them to fail is lack of adoption. Agree with that? Okay, cool.
That's why we're talking about it. So what are the barriers to adoption? Why does uh things that we why why does why do things not get adopted? Maybe we could integrate.
Integration difficulty into existing systems like that, yeah. What else? You gave them the choice. Oh, you know, I love this sort of Stalin-esque way of thinking is I agree entirely.
One person makes a decision, what's a decision? You do this. I agree. That's a decision. We've collected it with I've I've collectively made a decision that you're doing this.
Beautiful. Yeah. Uh what else? What's the other barriers to adoption? Quality? Yeah, so quality can that there's two kind of sense of that, right?
There could be trust. There could be just lack of integration opportunities, right? Uh what else? Uh, what does trust really mean here? So in other words, um, how does how is trust the barrier to adoption?
Why is give someone give me some detail about why trust is a variable, like low trust is a variable adoption? So data quality is one of them, right?
Because you might just not trust the data with data. What else? That it will save time because you can't see how it's going to be used or like envision yourself using it either.

Good. Yep. So trust that the payoffs aren't worth aren't worth it, yeah. Um cool. That there's liability on you for using the system. Right? So one way we deal with fears of being liable for things is we have again lead with leadership that serves that function.
If a leader tells you to do something, then you're no longer liable for doing it, right? That's when the value of the things I often say is do the thing I've told you to do.
Oh, but I don't want to do that. I want to have to think for myself. Well, what if I get told I'm doing something wrong, right? I just say, blame me.
It's fine, just go and blame me. I don't care. No one's like just move on. Just get just do something, please. Um, so uh great. That's a functional leadership.
Uh what other what other barriers to adoption are there? Got low trust, got low trust potentially we don't want to use it because it has low quality, or we don't use it because we don't think the payoffs are there.
What other barriers are there? Computing training. Lack of training, good, I like that. Like a training, yes. What else? Awareness. Not even aware of the thing, are we?
No. Competing priorities. Competing priorities like that. Well, why don't people adopt things? Why don't you adopt things? Have you been asked to adopt something and didn't?
Yes. That has happened to all of you. I assume. Right. Why didn't you adopt it? Compatibility. Pardon? Compatibility. Just constantly like, yeah.
So resistance to change? Resistance to change. Didn't actually see the value. Didn't see the value. Didn't see the value versus you learning something. Because you accrue a cost to learning something, right? Which is you have to spend the effort and you have to retrain yourself and you have to do all these different things.
Like, well, you know, I'm I'm kicking three buttons. You've shown me one button, but I know three buttons work, and I can't be bothered thinking about one button. Right? Yes. Um so good. What else? So that's a that's related to safety. I feel safe

if I'm doing this. Most of these things actually come down to safety, things and safety.
I feel safe for these reasons. I feel safe if the data I can trust the equality, I feel safe if uh I know there's a payoff, I feel safe, etc. Um what else?
Any others? Awareness, it's not safety issues, just awares. Yeah. What other problems? If people haven't adopted something, why haven't they adopted it?
Doesn't support the use case, yeah. Miss you've been missold. Yeah.
Say more. I guess like, you know, everyone has to go through like planning and you're planning, you know, you dedicate a certain amount of time to an effort, but then if it's gonna take more time to set up a new like system or a product, you know, you're less inclined to do it.
Cool. Alright. How do we do with any how now next question? Well, that question subpar if you like if you want a heading structure. Uh what how do we deal with those issues?
So one of the words I was using frequently there is the word leadership. So leadership is a big solution to those issues, which is I'm gonna I'm gonna reduce your liability for doing this.
You blame me. Um so you feel more comfortable experimenting and trying it out, right? And I'm gonna take the I'm gonna take lead of making people aware of these things.
In other words, I'm the nominated person to evangelize these things, I've got it out. So in some sense, leadership is a big part of the story. But let's be more specific.
How do we solve some of these uh adoption issues? Do we show that value of envisioning product references product? Good. So show the value. So rather than just speaking abstractly about potential benefits, you literally just show someone something.
Um yeah, I mean that that has been my experience in life. If I if I've you know I say that do it this way, and someone says, oh why do I do it that way with all these benefits, right?

And then you come back a month later not doing that way. You then eventually, after like four or five times, you've always leveled increases, start screaming at them, why are you not doing this, right?
Then you say, give me ten minutes, watch this Zoom call, boop, bop, boop, but don't, and then now they go and do it, right? Because there's just some blocker to I don't know, what is it?
Um seeing the value. Don't see the value. What else? Training, yeah. Arranged training. Good. Well email campaigns, communication campaigns, internal announcements, boards.
What about uh safety? What about this sort of like don't feel comfortable doing something? I guess with demonstrations with bribe and threaten people, right? Just give some reward if you do it. There's some punishment if you don't do it. But of course, if you're doing with clients and consumers and people outside the businesses are hard to think to do, maybe.
Deadline? Headline's good, yep. Because it forces prioritization. Anything else? A lot of training with documentation. Documentation, good. So once you've done your Zoom call where you're showing some someone, recall the zoom call, put that on the thing in the future for people to watch.
Tell people to watch it. Don't waste people's time. I mean, part of the background of many theories of adoption, of course, isn't simply a narrow issue about did we have a campaign or something.
People may not not trust the source of the thing, right? So it's about, of course, um being reliable in your communications, right? In all the things you've been talking about, you have to have a certain level of reliability.
So if you're if that reliability drops below at a certain point, you'll just won't trust that you're gonna honor your commitments or do things in.
Alright, so some of the things we've been talking about now are in the design phase of a data product, if you like, which uh when I say design phase, I mean the very beginning of the design phase, which is called in the paper, the triage phase or something.

But it's about um the conversations we have before we even really think about the the full technical details or whatever. And it's cool. Um right.
Alright. So there's lots of different things we can talk about now. I'm gonna have lunch at 12. We'll come back at one, maybe. So sometimes I mean that's mostly probably what we'll do for lunch in terms of times.
Sometimes we could go a little earlier, maybe like can you if we get done talking? Okay. Go on to the next topic. So there is there is another thing we can do here, which is a group challenge around an interview.
I might do we might do that tomorrow. Just so you're not grouped challenged out. What we might do as well tomorrow or this afternoon, is we might rotate the tables around a little bit.
Just so that people have a different experience. If you hate the person you sat next to, like you, for example, I can see the hatred. Hate presentments, business, spice.
I mean, you're planning, aren't you? You're thinking, how do how do I get back at these people for being awake? Um, of course not. But if you want a uh different experience, uh different people or different teams, etc.
Uh, we'll mix around, maybe you have to lunch for you tomorrow. Um, okay, let's talk about let's let's move on then. So as you move into so and as you answer some of these questions formally on a piece of paper or in a mapdown document, um, we might create a product requirements document, PRD.
Uh we might create a data contract, uh, which is a more specific kind of requirement might be right. Now, in order to create these kind of documents, we could create them by hand.
Remember this, remember typing things. Do you remember that? All the way back in January that you would you would type something on a computer, remember that?
Not anymore, right? Not anymore. Now you scream at an AR, don't you? Now you scream at an AI, and it does most of the tapping for you. And if you're really advanced, you might even just put transcriptions down.

You might even use a use a microphone and just talk to the system, right? Um, okay. Um now in CG, if we were to use AI to help us produce these documents, what would we call what would we call a system?
Anyone know? Begins with CPUs, if we have DPLC, but what's part of that? What's that part of? Yeah, catalyst. Do you all know about catalysts? Ever seen catalyst?
No, good. Right. So catalyst is a series of AI skills provided, I think, by Edo for uh for creating um everything you need across the data product lifecycle.
So design documentation all the way into the technical artifacts, Python and EBT and all these other things that you might want to create. So catalyst is an AI skill. We'll talk about what that is now. So did you set up catalyst as part of your onboarding? Yeah. Some people nodding, some people not. What we'll do is before we go for lunch, I'll ask everyone to type something into catalyst, and we'll see if it works.
If it doesn't, then you know, over lunch you should set up your catalyst. Go and find the onboarding documentation. Do the missing research now.
Um, okay, so let's talk about AI and AI skills. Um AI and AI skills. So I want to have a focused conversation before lunch. We can have a broader conversation after lunch, I suppose, about effective AI use and how we use AI effectively and what's going on with AI.
We do that to lunch. Before lunch, let's be a bit more focused. Uh what are uh so what is catalyst for? What is catalyst? What are AI skills? What are AI skills?
So when I say skills, do they want to what agenda AI is? Catalyst is AI skills. Oh, okay, great. Thank you. A repeatable set of instructions. I don't expect you to know the answer, but I'm curious if some people do, right?
So if I show you catalyst, uh GitHub on it on GitHub. If you have a look at some of these repeat what you say in repeatable instructions, they're markdown files with prompts in them, aren't they?
So if we say uh data contract, for example, and we look at the data contract skill, what we find is some text that is a prompt, if you like a saved prompt that's been

created by you do.
And this prompt um has this contents, it says, go and produce number three here, produce a structured document that matches a template below. What's a template below?
It has identity schema, SLAs, data quality, etc. Cool. So if I use the data cut uh the data contract skill, if I use that skill, what should happen is the AI should generate um a document that matches this structure.
Cool. So one of the values of having these skills, which are basically just saved historical prompts, basically, just save prompts. Uh one of the benefits is that when we create a design documentation or other artifacts, they're aligned with CG policy, they're aligned with common standards, etc.
So we're all doing the same. Cool. Now there's lots of lots of catalyst skills. Let's have a look in the main uh list of skills. So we've got um data contract.
We've got databricks stuff, dbt unit test. We've got DL DPLC. What does DPLC stand for? Data product lifecycle. Data product lifecycle, yeah. We've got DP DPLC concierge.
A phrase that no one can say. Right? That's Edo for you, right? What should we call this? We should call it DPLC concierge. What is DP, what is D PLC concierge? What do you think that does? It takes a long time to run, so no one knows. It does the entire DPLC. It goes through the entire data product lifecycle.
So if you start that on a Friday morning, maybe by Monday you might have a tape product, right? It takes a long time to run. Um what I would recommend when we're working with catalysts, as we'll see as we go as a course go through, is using more specific catalyst skills, because some of these larger ones take a long time to run.
Okay, but we but for every phase of the lifecycle, we have a different uh lifecycle skill. If we look at this, I'll just show you what one of them is, for example.
If you go into design, which is what we're talking about now, look at this skill file, and go down, and we can say, okay, go and do all go and run all of these skills in a sequence.

So go and run the requirements acceptance criteria skill. Go and run the feasibility report. Go and run the consumer use case back. Go and run the data modeling.
See this? So these lifecycle skills, they go and call out to these some of these more specific skills. So let me show you what does that mean? It means they go and include the prompts, those prompts.
So if you look at some of those more specific skills, so for example, requirements, acceptance criteria. Uh, what we would see is that when we run the design skill, the first thing we're doing is we're running in fact this skill. Which does more and more and more calling out. Cool. Skill calling. Cool. So catalyst is a long list of um pre written prompts that are useful for building data products.
It's more than that, though. It's more than just a pre-written prompts. You've also got some Python code. So for example, if we do maybe we do databrix search databricks somewhere.
Okay, let's put some examples in. Oh look at this. So there's some skills that come with Python code. And the Python code allows catalysts, if you like, to run programs.
So some of those programs can be searching data bricks for data products. It could be um running some jobs, finding some stuff. Cool. So catalyst is not just a series of pre-written prompts.
It's also provided script that allow catalyst, if you like the system. To go and search for things, create things, do things. Cool. Now this seems a bit mysterious to you.
Let's try using one of the skills. So if I go into my VS Code and I create a new chat here, new chat. Product requirements document. If I press forward slash, hopefully you'll have this hint.
And the hint will show you what skills are available under catalyst. So if I type if I type forward slash, uh, so that's just forward slash. If I type AI, I get a uh list of things, yeah.catalyst, and I can say product requirements document somewhere.

There you go. So if I do that product requirements document, and I go into my transcript, for example, that I've been recording this morning, and I copy and paste from that transcript one of your lovely group debriefs.
I could drop it into there, and we could get trying to get a product requirements work out here. Cool. Now since my transcript is on that laptop rather than this laptop.
That hasn't I don't have enough data for that. So let's do it, but we can we can maybe do um just copy that. See what we get.
Cool. Now if catalyst if we're actually using catalyst rather than just general general AI, at some point it will say reading reading a skill file.
Which it hasn't done here. Uh because I don't have catalysts in this uh I don't have catalysts in it. I'm not sure I have catalyst in this. Um have it in another one.
So over here I should be able to say AI catalyst product requirements document.
We go paste that doesn't okay. Um could be working, could be working.
Could be working. I'll just I'll double check it as working. So what I want you to do now is I want you to do that. So I want you to try creating a product requirements document with catalyst.
So what you need to do there, and I'll can leave this on the screen for you if you like, uh, is into your uh chat window on the right hand side, you should say something like uh forward slash AI catalyst.
Product requirements documents. And you should give um each of you should give a description of a data product that you've been talking about. And I want you to work in pairs, and what you want you to do in pairs is do all the same data product, but hide from each other what you're writing about, and then at the end compare the contract the requirements documents.
Cool. And the point of that exercise is to show you uh is give you an idea of how fragile the system is or how specific how how uh sensitive it is to how you're specifying the requirements.
So the challenge there is challenge and pairs. Challenge pairs.
If you haven't downloaded it, just work with the other guy, work with some some

someone set next to you too. Okay. We've downloaded it over lunch.
Uh-huh. I had a little bit of a needle of the slide group. I got to slide. When you're done typing. Uh what does the ITG stand for? Shall I use this?
It's a tending it's a um team within groups. Shall I use this for getting the access?
Yeah, um, this is a separate section to the access central.
But maybe this is more relevant. It wouldn't be heard. You shouldn't be needed if you if you are you literally unable to go to if you go to the link in the thing, can you not even see the repo?
No, it just says not find. Ah, that's a bigger issue then. Yeah, yeah, yeah. So that's with that. And that should be the right player service request for IPT is fine.
I guess so. I um delivering it. That's why the screen is. You could have two phones. I've got a new one at the weekend. So you can see 17. Uh 18.
Yeah, so you can play highly consistent. So you have like a user. What group does this need to be assigned to I don't know. Everyone is doing is there's a different?
I think we're supposed to be doing this. Access admin? Yes. Okay. But you need to give it some input. And just put it in my um. Data engineering, maybe.
I didn't do any of this myself. It was all done for me, so I didn't. Uh why was it not? Done for me.
Yeah, I got like a lot of both. Oh yes, and catalyst. Should I uh download that as well? That'll be part of the what you do. So you've got linked so lines.
Yeah. I'm in the confluence, by the way. Let me know if there's anything I need to look at in there.
Um, but like but the thing is you're giving it to people if they do the product well, we just do what the final product by normal is an SPR stands for.
This was the weather looking. I don't know, I think it's like the perfect temperature. Like twenties rather than pushing forty. Right. Yeah, so yeah, like yeah, I do know just like either B but link or you know how to uh these are back to those.
I gotta put it back into my Docker. It's gonna be like you shouldn't make it, you've

got to calibrate. Oh that's Mac. I wouldn't need to say to me if they have one. Yeah, yeah. I don't know. Yeah, I think there's a lot of this. I don't like it.
Well my computer's gonna do an update. In two hours. Yeah. I'm not saying just do it now. Since we're going for that. Oh yeah, I always find that.
I can't record it. No, it's just gonna be a good thing. Yeah, all social. All right, so yeah, that didn't bring that books and you have a US phone number.
But I can make calls.
I see why you're putting this way if you didn't know. Actually, I think that one about it.
And if the transcript isn't on, it's now your fault. Um, two, one, two. No, okay, that's not working. One, two, one, two. There we are. So uh the hours for the course are nine till three, nine until three with lunch at twelve, break in the morning, break in the afternoon.
Um the capstone event on Friday. Um runs till about two. So hopefully get off a little bit earlier. That's if you are focused and determined. If you're not, then we'll be here forever.
Um, okay, so uh let's talk about AI. As I said, after lunch, what we'll do uh AI and effective AI practice, I suppose. Let's talk a little bit about the lights where I would where I was standing.
Um let's talk a little bit about the difference between AI and catalyst, what what the skills are, harnesses, how the model is created a little bit, and then some attacking practice stuff.
Cool. Um so where should we start? What is AI? Artificial good. I like I like how people laugh at the absurdity of answering questions by giving me the initials or the initials.
Everyone else just gives me initials and no one realizes how absurd it is just to repeat the initials. I like that, artificial intelligence. That suggests this room is ahead of the others.
You realize how bad your answers are. Um which is a step forward. Yeah, so let me for your own sake, just for a casual interest, let me give you my definitions of

AI and other things.
And I can play by computer scientists. Computer scientists, you should never trust them. If you're a computer scientist walks by, make sure you've still got your wallet in your pocket and you're not and no money missing or anything because you know they're always on a scam these computer scientists, something.
So my definition is that uh artificial intelligence is a kind of research project or the or a human project of creating a system which can perform tasks that in human beings require intelligence.
So it's reference to tasks that human beings do, and a subset of those require intelligence, and artificial intelligence is a project of building artificial systems to do that, right?
Now, uh intelligence. Let me define that for you. By the way, all of this is transcribed, it's on the notes as well. So if you uh feel like it would be nice to have these, they're in the transcript.
Cool. Um I want to define intelligence, unusually, maybe, as a strategy in animals for coping with a lack of solution. Strategy animals use to cope with lack of solution.
What that means, and I can tell you what the what's the strategies are if we're interested. Don't know how interesting these things we are, I'm gonna move forward, but if you're curious, I can give you some.
Anyway. So uh a strategy in animals for coping with a lack of solution. What do we mean by that? Well, if you have a solution to a problem, then my argument would be you wouldn't need to use intelligence to solve it.
As soon as you have a solution to a known problem, you can automate that solution. And I would argue that any automated solution doesn't really use intelligence.
Now that's the opposite view of computer science. I'll explain why in a second.
Now what that means is what that means is that um, yeah, as soon as we find solutions problems, we we automate them.
Okay, uh what else? What else did it say about that? Let me tell you a little bit how

uh what's right to animal use, and I'll sort of play by the computer scientist definition.
So, what strategies do we use when we don't have a solution? Well, there are many problems in the world in which, for example, two animals are adversaries of each other, cat and mouse, and there isn't a kind of mouse behavior that is pre-programmed.
So the cat can't say the mouse is going to do A, B, and C. So I'm not gonna go left or right, but the cat doesn't know, right? So there is no generalized solution to the problem of how do you hunt a mouse?
Because the mouse itself is not the kind of thing which is programmed in that way, right? So the cat has to adopt a strategy, which is how do I use my techniques, how do I move around, how do I explore, how do I use curiosity, how do I use sensation, how do I use creativity, etc.
Many other things, imagination, primarily. Um how do I what I'm gonna imagine the mouse gonna go in this direction? And the cat imagines that before the mouse does that in the other direction.
I mean, we actually know that the map that the cat does that, so I mean we can actually look at rats' brains and see them imagining the future before they actually do things, actually see the uh planning taking place.
So animals use imagination put principally. But the other things to plan uh to plan out of the future when things are uncertain. So um, okay. What is it to say then? So artificial intelligence is so artificial intelligence is looking at those tasks that human beings do where we use intelligence because we don't know the solution, and it's the activity of trying to find, okay, are there a space of solutions within that area where we use intelligence, can we automate those?
Cool. So computer scientists have the opposite view. Let me tell you why computer scientists are the opposite view. So if you look at a human being trying to solve a problem, you can see from two sides, right?
You can see from the side of what's a problem, what's the solution, so that's the formal side. Or you can see from kind of the inside of the head of the human

being, right?
Now if you see it from the outside, you go, there's a problem. Creating a data contract or whatever, right? There's a solution, there's a solution, right? Computer scientists will go, well, I'm not interested in the human being or the animal. I'm interested in the problem solution. And so the computer scientists say defines intelligence as simply that we have produced the solution.
Make sense? And you can see why that may be.
Anyway, two different views. Now the virtue of my uh one of the uh sort of virtues, if you like of my view, which you might call it biological view, uh, which is that as soon as the computer scientists have solved the problem, it's no longer it's no longer using television.
No longer problem. Yeah, exactly. Um so as you so anyway, anyway, now why are we talking about this? Okay, so when we look at uh solutions to problems, so for example, uh general generative AI and things, we're always looking at what I would call a proxy model of intelligence.
Now, proxy model, what proxy model is, what we're doing here. Uh proxy model means that we're we're looking at a system which is trying to uh trying to acquire some of the properties that the human being has uh with actually actually having them.
Make sense? Right? So in the case of generative AI, which is what we're here to talk about, just in just throwing these things in uh bit of intro.
Generative AI. We'll talk about LLMs and the way we'll talk about that in a second, but the way that uh when we're dealing with copilot or whatever we're dealing with in VS Code, what we're dealing with is a operates in what I call the text domain, which means that we have words going in, words going out.
And so if we want to, for example, imagine the future, we have to emit a m a certain amount of text that describes the future, right? We don't have any direct capacity to imagine the future.
Does that make sense? Like the LM does not have uh like copilot does not have in it in itself any ability to imagine the future, there's no ability to simulate.

Just telling you that's a fact. So if you think, do LLMs simulate things in the future, the answer is no. What they do is they generate text. If that text has a description of the future in it, then in some sense it's a bit like it's in making the future.
But it kind of hasn't. It kind of hasn't. Simulated one or two possibilities. Make sense? I'm saying, huh? Okay, so generative AI. So it's a topic about two types of AI.
So there's two types of AI. There's discriminative AI, discriminative AI, and generative AI. And we can define these somewhat formally. But discriminative AI is wherever we're labeling something or attributing something.
Making sense? Now, for example, descriptive AI is whatever we're giving a prediction or a label or a classification to something, and the value, it's kind of an answer to a question.
Think of it that way. Descriminative AI is your answering question. What's the price in the future? What kind of thing is this? Yes or no? Sense?
One way thing about the answer. Generative AI is wherever we're generating something similar to the things that we've seen before. So we're not really asking questions.
We're just saying, here's some pictures of cats. Give me another picture of a cat similar to the pictures of cats we've seen before. Now, in the case of LMs, which are the AI that we're dealing with here, code, etc., they're generating text documents similar to text documents they've seen before.
Cool. Okay, so let's talk about LMs. What is LLM stand for? Large language model good, large language model. Large language, and blah blah blah.
So how are how are LLMs created?
How do you create train? An LLM. We do have data sets of these are we need some of the existing data sets.
In the case of large language models, what is the the name of a historical text data set called the corpus? Anyone know what the corpus for an LLM is?
What's a major part of the historical data for creating an LM? The internet. The

internet, yeah, exactly. The worst possible thing. So why why why I'm just bickering with you all day about nothing?
We'll talk about that. So I've got the internet. What else do we have in the historical record? We have all ebooks. Everything ever digitized, everything ever all electronic text.
Now in modern LLMs, they're provided by vendors. The problem the vendors have, or could have, you might imagine, is that they some years ago, they they'd already have all the text ever written, right?
So now if they went to the internet now and they found new text on the internet, it's quite likely that new text would have been produced by an LM.
There's a big problem with that, which is that LLMs are not like animals. This is one of the characteristics. If you say, well, well, how would you tell the difference between a system which is genuinely intelligent one which isn't, right?
Here's one way of doing it. So if you feed the output to the system which is genuinely intelligent into a system into itself, right? In the words, I look at my hand and I look at where my hand is, that tends to stabilize the system.
Right? So in animals, we have a hand-eye coordination. We have an internal sense of ourselves. When we when we feed our own, when we feed ourselves to ourselves, we actually stabilize.
Because we sense of how we get a sense of how wrong we are, right? Let's call control. You want to go to that with call control. We can go down that route.
LM's have exactly the opposite way of working. If you feed the output to an LM in its own training, it will collapse in its ability in its performance.
Now the there's lots of reasons behind that, but basically the arrows explode. So you think of it, if you think about what an LM is doing with all this historical data, what it's actually doing is kind of compressing it.
So mostly what an LLM is, is a very simple compression of the historical data. So you think about compression, you take a JPEG, you compress a JPEG, it gets blurry, right?
Keep compressing and keep compressing and keeping it, it's blurry and blurring

blurring, almost nothing in it. Make sense? So the process of creating a statistical AI model is a process of compression and compression can solve data.
And if you keep compressing things on the output of compression, you end up just spiraling away. Now we can again if we're only going to talk about these theoretical things for like 15 minutes or something, just have a bit, you know, kind of interest stuff, but if we were had a more dedicated amount of time and could explain how that isn't.
Why that didn't happen at all? Any case. Um so what else did you say about this? So vendors have a problem, which is they've already juiced, or they're basically compressed the entire internet and everything ever in.
And all that if they if they if they took any more of that, it would just end up making it worse. So there's one additional source of new data that's very important for vendors that is definitely guaranteed a source of human data. Does anyone have an idea of what that is? So where in the entire internet today can we guarantee that there's a source of true, let's say low error human being data signal.
Where do we get that from? There's a small number of places that produce genuine at large scale produce human data. Could be. Have you met any human beings in social networks?
Uh no, could be. It's the AI vendors themselves. Because you're sending them a huge amounts of track messages. Right? So the on mass, they're the ones.
You know how I said finance industry exists because it sits on top of the value stream of the thing. So one way AI vendors can produce frontier models, is they sit on the value stream, human being value stream, right, on the internet, which is the source of human data, and they sit on that like uh like a troll on a bridge or something, and they watch human beings send the prompts.
So you're extracting all the value from those problems, right? So now we have internet, we have e-books, but we also have, we can call it like all the chat for all the all the vendor data, which is massive amounts of text, right?
Now this is quite useful because if an LLM isn't very good at something, you fix

it. Because you keep typing. Then they can observe that typing, and then they're gonna add that to the LLM.
So the LM appears to get more intelligent, appears to get more intelligent, but really there's a collaborative human effort, so it can essentially laundering that intelligence through human beings, right?
So collaborative human effort to provide massive amounts of training data through these problems. Now, okay. There's a few other kinds of data we give the LLM.
So the LLM uh generates text, it does so based on the similarity of previous text. However, if you do that, if you just generate it similar to previous text, isn't very useful.
So that's the state of LM, the state of large language models in around, I would say 2020, 2020. Um not that useful, right? Now uh we have to do additional things.
Anyone know what the additional things are? Some of them might know. Might not. We want to call instruction fine-tuning. Direction fine-tuning. Or uh ISF.
The other one's called reinforcement learning routines. So instruction fine tuning, what instruction fine tuning is we go to Kenya, we go to India, we go to Nigeria, and we pay 10,000 people to produce data sets for us, customized data sets, that show an instruction and the machine correctly following that instruction.
So we say write a poem and we write a poem. Cool? So those are artificial data sets. We add those to the mixture. What they do, it's called instruction fine-tuning.
What they do is they emphasize the instruction words, so the LM doesn't just generate text similar to previous text. It focuses on the instruction words.
So it makes it use it makes it kind of more like a system. Make sense? Cool. Um the other thing we do is called reinforcement learning with human feedback.
Now that's not, that's that's quite different than the other ones. This is where we hire experts. Uh we hire large enough experts. We might hire you thousands of mathematics PhDs.

And we say, given that you've generated this result, I've asked you to generate this thing. Here's the thing you've generated. Generate the three or four or five different examples.
I as an expert want to say this one is better than that one. It's called preference, right? Preference data. Now there's one area, there's one area where where we can automate this preferring of outputs.
In other words, we can automate the idea almost like having an expert that can say we prefer this output versus that output. Hard question about ask it.
Can you have an idea where that area would be? So what kind of thing can we ask an LM to generate where there's an automated way of finding out the thing it's generated is of good quality?
So for example, poetry would not be a good example of that. Mathematics, kind of, yeah, because you could maybe have a theorem prover. But the reason mathematics works at all is because it's written in code.
So it's code, so it's programming. So the reason LLMs are particularly good at programming, is you can have a programming compiler, interpreter, whatever you have Python or whatever, run the code, and say that works, that didn't work.
That's actually quite a weak quality signal, because code working is not so great, right? So we have to do additional things. But the reason LM is a particular bit of code is because you know you can have trillions of code examples generated, and you can have the in an automated way, so that will move them with that.
Right. So uh this is yeah, so in today's world, we have um probably a half the time in creating LM is creating it from this corpus data, and half the time in creating an LM is actually creating it from this font we call fine-tuning data, preference data interesting.
How much did it take to how much does it cost to create an LM? So let's say on the uh I would say on the order of, yeah, on the order of 50 million, maybe.
Yeah. Um 50 million. So within some factor of 10. Yeah, like two, two, three. Um, how much does it cost to run an LM? Uh front, frontier event. I mean like a frontier model.

Oh, okay. Because it doesn't cost that much to run a to create local one If you're creating a small local one cost millions of right. Uh what did it cost to run ChatGPT thing?
Per month. I have a guess. Yeah, so on the order of like a hundred million a month. I mean, we know from anthropics leak S1 filing. Um it's what, eight billion a year or something, so it's divided by something, right?
So a few hundred million a month. So, where does the cost uh for running LMs go? Wow session. Where does the frontier model cost go? It doesn't cost it doesn't go in training LM.
Training LM is one less than a month of cost, right? It goes in running uh an LM. Now this is not where people thought it was going to be. So go back in history, people thought the moat was going to be that very few people could um train an LM.
The moat now is actually very few people can run an LM. So this is why China is not frontier at the moment. It's actually not because of the training uh office goals, because it doesn't have enough um infrastructure for that.
But that will probably that'll probably change. Okay. Um let me say something about why this is, because that's gonna lead us in to talk about catalyst.
So this difference here is talk about catalyst. How does it do that? Magic. We'll see. So why does it cost so much to run an LM? Power. Oh, yeah, yeah, yeah. Now uh power is running at so that's a yeah, that's correct. It's just not the answer I want. Uh shall I just tell you? GPUs, yeah, these are all correct. Compute. Compute. What are we computing? Yeah, computing the million of cloudition or instruction given by the GPU and something for the word. Yeah, so if we have an LM, right?
What LM is going to do, is going to take the entire conversation history that you've had so far with the R, the entire history, and it's going to predict the next word, yeah?
Yes, in the right predemption, right? It's going to continue the conversation. That's what an LM does. Um so every time you add a word to a conversation, that

goes back in as the input.
So it's always giving the entire conversation in as the input. Cool. Now, um this process of just predicting the next word, I'm just giving you one more word, that's quite cheap.
So if I ask for a word, the next word, or even a thousand words from an LM, it's very cheap. Much cheaper than the actual running costs. So where does running costs come from?
They come from behind the scenes in modern LLMs, they come from the vendor generating a large amount of tokens for you behind the scenes. So you put your conversation in, right?
You get your response, which is called completion, by the way, if you want to know completion. And it seems like, well, I've got my completion from this conversation.
A modern vendor like Althropic ChatGPT or ChatGPT or whatever, um, they're actually generating a large amount of reasoning. Uh given the game away. Maybe they'll tie today, I don't know. Let's say what is that what is that called where they generate a large amount of tokens behind the scenes?
What's that called? Token? Nope, gives it R. Inference. Inference, the whole process called, yeah, the process of running an LM is called inference.
Yeah. Training is making one, inference is running one. Right. Begins with our reasoning. Easily. So it's called reasoning. The reason it's called reasoning is, well, it looks like reasoning.
So what they what they do is when they create LLMs, um, they kind of divide the training. Basically the way the training LMs is to produce a long conversation. Make sense? So they're training an LM with this prompt produce this conversation. Now, as part of that conversation, what they what the trainer wants to observe is a large amount of intermediate tokens between the question and the answer.
Large amount of intermediate words. So for example, if you ask the question what's two plus two, they don't want to go all the way to four, straight to four.

They want to see the user uh would like to ask a math micro question. They're adding, they're asking about the number two added to the number two, that's four.
Make sense? So every question they want an intermediate amount of tokens, and the reason for that is if they jump straight to asking what the next token is, that is uh that is a lower quality response, because whenever you just add, whenever you put one word at the end of something, you're always just generating something similar to the things before.
So if you say what's two plus two, all you're ever really saying is given all of the things you've ever seen, what's the most similar word next to what's two plus two?
The answer before. But if you say, generate my generate me a data contract with American mutual funds, blah blah blah, right? There is no similar conversation. So if we just ask what the next word is, there is nothing similar. So you might get garbage. But if we put if you put a large amount of intermediate word between them, where we say, for example, this is a CG investor portfolio, this is what CG does.
Let me go and search the web to download this other thing and put that in the middle as well, get all the way down, suddenly then the next word out, even though there's similar to nothing before, it's still on average now, it's meant to be a bit better.
Right. So reasoning, this this process of reasoning is where the vendors will on your behalf generate a very large amount of intermediate words for your tokens. And now it's all makes it cost money. So in the history of LLMs, we go, okay, we think we think training LLM is going to cost a lot of money, therefore we win.
Oh no, everyone can do that. Okay. So we we go to a model of reasoning, but we generate lots of tokens behind the scenes. That means it costs a lot of money to run them.
So we can go win because we can pay a lot of money, right? So how do we effectively uh choose what context will I assume the vendors say how the sound

amount of words look at too, but like this is bigger as LLMs like draw bigger type of DLP, potentially like paying like oh talk about like scheming.
I should love in these words, how do you do that? Do you know? Well, it's the secret source, what's the secret source? Uh I mean um some things can be said. So for example, part of the reason cybersecurity LM is very good at cybersecurity, is on the internet there's a lot of forums, in fact, a lot of uh examples of people in forums just discussing cybersecurity.
Like if you go to the sub-security conference, right? Let's have transcripts of a YouTube videos of 20 years of cybersecurity conferences, most of those talks are just reasoning choices.
Someone says, I hacked this by doing that, and then the rest of the talk of the hour is someone going, and here's the steps I took. Right? And like in mathematics, what are mathematics papers?
They're like prompt conclusion, here's all the here's all of the reasoning you need to get to the from the conclusion of the problem. So any cases where there's overrepresented already in our corpus, examples of people showing you they're working, the LM's are gonna be very good there.
In cases where there aren't there, they have to go and pay someone to actually produce that intermediate steps. So there's so the areas where the N3D steps exist, you're gonna get a reasonably good quality.
So cybersecurity is one of those areas, you might think, well, why is it cybersecurity code? Because everything in cybersecurity is the entire history of cybersecurity, has been from the very beginning, people just writing books on how what the steps are.
Explaining their reasoning. It comes from like that. Comes from the hacker culture of the 70s and it's it's kind of like a the the culture of cybersecurity as a practice is very much um kind of shared knowledge culture because of the nature of what cybersecurity is, you can't be very good at cybersecurity if you just keep everything to yourself.
So to be any good, you have to have this culture of sharing how things work.

Okay. Where are we going now? So there's one thing to add to reasoning, which makes a uh um which makes uh AI useful, is in the conversation traces, we then have the prompt, we have the reasoning, we have the response.
This is still all text. Yes? So this means that the LM can't do anything, just test. Right? So there's one thing we need to add to this trace, this conversation history. That's called a tool call. What that is is a specific kind of text that the LM will generate for something else to interpret, that means go and run some command on my behalf.
Right? So when we're training an LM, if I say go and for example, write me Python script to do something, what we really want to see is here's my prompt.
Here's the hidden reasoning. Here's a tool call, called tool call, and then here is the actual code down here. Right? And what would a tool call do?
It may check my Python version. It may see if there's any files relevant to the prompt that I've given it. Cool. So when I'm training an LM, I'm training it to produce conversations that have all these parts.
I'm not just training on like the internet. That's a that's like five pennies or whatever, right? I'm not just saying go and go and copy this e-book.
I'm saying this is the exact structure of the thing that you should be producing.
Now the tool calls, this is just if you know what this is, just JSON.
Just JSON. It's labeled tool call, basically. Part of the computation labeled tool.
Now, what interprets these tool calls? Hey? H GTP? Could be.
Oh. Yeah, of course. Well we remember that. You know, we were in, we were like eight years old, we were doing the the you know the salt put salt in the tray, it evaporates, leaves.
Then they also said at the same time, you weren't paying attention. They said that the program that interprets an LM's tool call is a harness. You've all forgotten that.
Um so a harness, so in our case, the harness is VS code, or it's also copilot. If you've used from if you've used AI from the command line, none of you have probably, maybe not all of you.

Um, clawed is the harness. The model is called opus, right? Claude is the harness. Now, in the modern era of AI, you know what we're gonna talk about? I training was the big thing. Then it was reasoning, inference was a big thing. Now it's harnesses other people. In other words, the big thing is what we call a genetic AI, which is where you use an LM to generate tool calls for you, so it does things for you.

Right? That's like the third wave of AI. We're in the third wave. AI. That's why we talk about catalysts while we talk about skills. We're not asking it to generate similar things, we're asking it to like reason for us, we're asking you to do do things for us, right?

Yeah. Then what is the agent? There you go. So an agent, we're gonna define an agent. Remember computer scientists are always trying to where's my money gone, right?

What I did like part copilate all if I agree. Oh, yes. Yes. So look. So an agent, there's a harness plus an LM. So what does that mean? It means if I say to Catalyst, search Databricks Marketplace for a data set that is XYZ, right?

That's a piece of text. That goes to a vendor, LLM, and it replies with behind the scenes lots of reasoning. And maybe show you a preview, a summary of what the reasoning was.

So in your VS code, you can sort of see it tick by. That's not the real reasoning, because that's hidden from you. Because this is proprietary. So they show you a summary of what it's doing.

Then it goes into a tool call. In other words, it's just generating text. That text is interpreted by the harness. In this case, the harness would be let's say ESL code. So this is a program that can interpret LLM output and do actions on behalf of that output. And this is a program written by human beings to do that.

That's because LLM's okay, it only generates text. You can't do anything. So once you have a harness and you've got an LM coupled together, you have got an agent, because the LLM is generating actions and the agents performing the actions.

Now when you have catalyst, what catalyst then is it's prompts given to the LLM, which may generate text, but also may generate tool calls, may do things for you. Cool. Right. We call those skills. So skills are prompts, say prompts that you give to an LM, which in general trigger it to run commands. Cool.
Any questions? That's our half hour on AR. I'm gonna show you some. I'm gonna show you some options now. What's this? Why is this up there? Who's doing this to me?
Self-inflicted idea. Father, isn't it? I thought people have a great colouring. How do I?
There you go. I am the cause of and solution to all of my problems. Okay. So let's have a look at this interface here. So what part what is uh let's just describe some parts of this interface.
What is this? What is this part here? LM. That's the yellow lemon. That's the model we're using. Model or LLM. That's the model we're using. For opus, sonnet, haiku.
What's the difference between these options here? Less token of option. Maybe. Or that means. There again. Less token comes up shape. Less token, yeah, that's from a pricing.
There's a pricing difference, right? And in danger. Yeah, okay. Uh uh, from a like a technical difference is that the different sizes of model. So they've they're they're they're compressed more.
So opus is the least compressed, if you like, of everything. You get all that in fine-tuning, all the data, blah, blah, blah. Like that's the best quality JPEG.
Looking at an image, loads compression. Haiku is the mode compression. To point where Haiku45 is probably not very useful. Um it's quite a C model.
Uh anyway. Um so Opus is the highest, and Sonic's mini per size. Yeah. And same for uh ChatGBT. So we chat GPT we have lunar, soul, uh, and terror.
So again, Sol is the least compressed of those three. Lunar is the most, and terra is intermediate. Cool. Now, does that matter? Well, if you're providing massive amounts of context in your prompt, the quality of the model matters less.

So you can go for cheap. If you're providing less in the prompt, then the model has to make up for your vagueness, so it matters more. Cool. Now, whether a model is any good at what you're doing is what you call an empirical question, means just something you have to try out and see.
There is no theoretical answer to that question. Part of the reason is we don't know how they're trained, uh, to that level of fidelity. But also, uh, different vendors will prioritize different parts of the training.
So, for example, it's commonly thought that ChatGBT prioritizes the instruction volume, meaning that it's quite literal. You say go and do something or go and go and do that thing.
And people think that uh Claude or anthropic prioritize kind of the preferential part of the training. Where you can give it vague instructions, and maybe it'll do better guesses between different vague options.
That might change. Okay. What's this part? Well the thinking effort. Ah, no, no, no. That's a computer scientist comes to sell you something. Oh, yes, the LMs think, don't they?
They generate text, right? Which is which is a model of thinking if you're like deranged. So someone says, what's thinking like? And you go, it's like generating text, and someone goes, but this person must be deranged, right?
So thinking is nothing like generating text, but it's a little bit like that. Anyway, this is behind the scenes reasoning. So what that means is this is the amount of tokens it will generate behind the scenes for you.
So if you put this onto maximum, every time there's a gap in the conversation, behind the scenes is a massive amount of things we can generate for you.
Now, you might think, well, why don't we always have this on maximum? Actually, a lot of performance can actually be worse than maximum, and it can be on high.
So you're thinking like if this was thinking effort, how could that be? It's not thinking effort, right? It's not thinking effort. It's it's intermediate token generation.
And the reason maximum can be worse performance is if you ask this simple

question, and if it's forced to generate a large amount of text, that actually derails it.
If you say what's a zebra or something, striped animal, whatever. And if it goes, well, the user might be asking about animals, and then it goes off on like for pages and pages and pages, it might give you the wrong answer.
So maximum is not actually the best quality. People say medium or high is good standard, and then you play around and uh what's the context size?
Memory-ish, kinda, that's kind of true. How many more have you in one or two?
Yep, so it's it's the that's the so the c the entire conversation history.
So remember the last gives the entire competition history, including all reasoning that you can't see, including all tool calls, right? That goes into the LLM when you generate one word.
So, this is about 872,000 kilobytes. This is a byte is like a letter, roughly, so it's like it's like 800,000 letters. So that's about 100,000 words.
Cool? And you might think 100,000 words is a lot of words, but they're generating massive amounts of things behind the scenes that you can't see.
Yes? Question? Is the UN or a question? Before it checkpoints. Before it compresses, yeah. So we'll come back to that in a second. So 200,000 is quite small, but actually 800,000 isn't that big either.
So actually, I would say that most non-trivial tasks, most like not one-off tasks, often take about a million tokens. So 800,000 is something like about a task. Because once you've gone through the conversation history, you've generated this on Python, you've iterated on that, you've got the Python right, you've got the thing right, da da da, you're probably up into the hundreds of thousands.
Now, just because the window is this large does not mean performance is that good. Performance tends to degrade the longer your conversation. We're almost done, we're almost done.
There was something you were saying about what was that last thing?
Compression. Compression, yeah. Okay. So performance degrades, if you go on for long enough, at some point, towards the end of your window, the vendor or

VS Code will choose to compress your conversation down to a smaller amount. So if you've gone for 600,000 words, it'll try and reduce it down to like 50,000 words, and then it will keep hold of that. Better is to start a new conversation. Better at some point when you're halfway through, just say, summarize everything we've been doing in a file, and then copy and paste that into a new conversation rather than because when it chooses to when it chooses to like checkpoint or compression, kind of random.

It'll choose its best guess at work. Yeah. Which is length-based rather than hand off code. Yeah, rather than hand. If you go like, oh yeah, we've done that now, and you think, oh, well, let's continue.

Probably in your own mind you can go, no, let me just summarize what we've done, let me just start on the next part of the task. Now you say it's one final point.

What you'll find is that if you go on for long enough, you'll probably start swearing. WTF. On my reason for that, well, it might be two different reasons. One is it could just be higher sense. Another reason is that as you go down the content, as you fill out a context, um, the performance of the model drops, um, and as you compact things, the performance of the model drops, and you finally not doing the thing you want it to do.

So there's a trade-off between restarting something and exploring a wider space of options versus being anchored down something, and because it's context is full of prior conversational history, it's always anchored by that history.

So if I'm trying to solve one very specific problem, and there's lots of things we've been talking about, then is the next word is going to be similar to the conversations with all that stuff in it, so that's gonna be not very useful.

Right. Okay. So in terms of uh effective AI use for ourselves, we think about effective AI use, we're thinking about being skeptical of AI when the thing we're doing seems particularly novel, unlikely to be represented in the training data, right?

Am I in an area where I think I can just say do this and it's likely to work first time?

That kind of area is likely to be something where there's lots of examples of the number forms, right?
If we're not in that area, and we're often not in that area, then we need to be empirical. We need to sort of measure the actual performance of the system uh using different options, variety of different options and different techniques.
And then coming up with an internal model of how good this system is at our particular task set. I'll be different across different people. So one person drop might be well certified.
All right, there's some more things I'll say on another day.
So why don't we take fifteen minutes? And when we come back, we'll do some group charging. Cool. So I'll see you in fifteen minutes time.
So now we just need to do what's the word of the other link. Yeah, but we are we already did that. So now you're just going to do JPO off logging.
And then you should be able to do that. Yes, it's like solved, so you just go to CG for the CP thing. Yeah, it's a CD thing. And what we did is we added the repo to the plugin.
So it's fine, it's fine over the web. And now we're just talking about it.
Do you see it?
So there's like some side.
So basically enter.
Sometimes we have to optimize it.
Or you're not going to be a little bit more than a lot of different things.
It could be two. Could be two. I fear it might be two.
I think we can do it.
You don't like all the access to it. Yeah. I think you don't underneath it.
Yeah, that's the hyper.
Oh, it needs to see a lot. That's that's supposed to be an absolute fixed one.
Yeah. I think it's one of these.
But they're very open to the use of it. Yeah. I mean, not here. But they're just all features.

Yeah. Yeah. That's probably true. Find it like a lighter at first.
No, I guess at least they got a pop-up that's in. Should I try to reopen this in the container?
That there's supposed to be a container.
We all hear more or less. Yeah, okay. So what I want to do now is to have a look at combining catalyst, get a product design, and maybe it's some audio transcription.
So here's the activity. I'll put maybe some questions up on the screen in a second. For those who have a MacBook, you have Apple voice memos. Do you know about Apple voice memos?
Yes? Some of you do. So if you have a MacBook, you can type voice memo and you can press record like this. And then Apple, Tim, you know our friend Tim at Apple.
He'll record some of your voice. There you go. Press pause. Press done. And you can press the transcript button at the top there. It will show you the transcript.
You can also, on the uh left hand side here, is if you click off it and then click on it again. You can copy the transcript. Copy transcript. Right?
And if you do that, then it'll write, you know, it'll be you know good enough. Right. Now that isn't an amazing, it's free, so it's not an amazing transcript.
And I know that there are uh there's there's things internally here where you have voice things, but I don't know the details of that. I do know that you're allowed to use whatever app is on your laptop.
So by by inference, you must have approval to do that, because you have it, right? And they would have taken it away from you if you didn't. So that's my reasoning.
Um, here's what I'm going to suggest is that one of you who has a Mac uses Apple voice memo and you press record, and then you have a conversation about a data product.
And then you use that to generate a P I D document and a data contract. Cool?
So here's the conversation. The conversation is going to be an interview where

one of you role plays the data producer, one of you role plays the data consumer, and one of you role plays the data creator.
Cool. So creator, consumer, producer. So you'd have to choose something where the producer knows the data source quite well. The consumer knows what they want quite well, and the creator is there asking them questions.
Well, how do I get this from this source? How often does it is it how often do I get this, etc.? Does that make sense? Now, try and roleplay as close as you can.
But of course, if you break roll and you help each other out a little bit, that's okay.
Cool. Now what else is there to say? What questions do you ask when you're interviewing?
Well, um their questions. So this is the second phase of design, if you like, which is suppose we've configured the work this morning on the feasibility and viability and cost questions.
So we agree we should build it. This is what we should build, right? This is how do you answer what we should build? Well, we ask, we interview the producers and the consumers, and we figure out what they want.
Cool? So let me um let me just give you. Uh let me bring up the latest version of the courseware where the use questions live. And I'll drop this in the team's chat.
Yeah. And as I said, by uh by the end of play Friday, if not by Friday, what I do is I put the transcript from this session in there as well, so you have that.
Okay, so what page do we want? I believe we were on page 20, but I've I made some modifications this morning. Where are the questions? That's the line canvas.
Please don't have removed the questions. Um let me just interview questions.
There we go. So here's some questions. I'll put the I'll put these bigger.
I'll put this bigger in a second. But what decision do you make with the data? Uh how do you make a decision today? Who uses the result directly?
How is the source generated? What do these terms mean in your work? How do you recognize the result is usable? When do you read the result? Cool.
Now you can ask these questions sort of literally as part of the conversation, it's

part of the interview. And you can play around with additional questions that might be relevant to understanding the consumer's need.
Cool? So I want you to ask some of these questions and some of the questions of your own and figure out what what the data product should be. Cool.
Now what we'll do this time is we will um switch up the groups. So I will put that on the screen for you. So here's what the exercise is going to be.
I will tidy this up on the screen so it's easier for you to see. I'll put this on the screen. Uh you'll be in your threes. You'll Apple voice memos.
One of you will have an Apple voice memo recording. You'll do the interview, then you'll drop the transcript into Catalyst, and you'll say, give me a PRD, and then give me a data contract.
Cool? And we'll see how far that goes. Cool? Right. So that's kind of a workflow model for working with AI. It's not bad, not bad. If you're on Zoom, right, you're on Zoom or Teams, wherever you're on team, uh, you'll have a team transcript. So you can use that if you're on a cool. Alright, so let's do threes. How many people are in the room? One, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventy.
It's one more than we need. Oh well. So mining. Ene meeny mining. Mining. E me mining. Eenie. Mining. Ene. Meaning. Right. I've not completely wrong.
I've divided by I've done uh divide by three, rather than divide by six is what I should be five years. Oh, trat. Okay. Let's do that again. Very fun.
One, two, three, four, five. One, two, three, four, five. One, two, three, four, five, one, two. Cool. This table is one. That table is two. This table is three.
That table is four. That little there is five. Go away. I always do that as well, by the way. I always do that. Never mind. Never mind. After we just started the bond.
I know.
Now what I would suggest is you try and roleplay for a little bit until it breaks down. And if it breaks down, that's fine. But the idea is that you're making the so the data creator sits in the middle making a data product and then talking to the producer and the consumer to figure out what the data product can be.

Cool? Producer's a vendor, it's the upstream. Uh I mean consumers downstream and vendors like upstream uh the producer's upstream, right? Cool. Creators and the upside.
If that doesn't fit exactly onto your use case or something, just a little bit. And you're recording it, right? Yeah, we have the point.
Point to the right. What did you talk about? Save. Okay. Yeah. I feel like I asked you to use something. Right, yeah. Well, well, okay, was there any subtleties or interesting things that other people in the room might want to hear?
Um we had the same conversation, but I think came from a different perspective.
Brilliant. Um more from a like a business user's perspective on some of those, so not super technical.
Cool. Um I think one of the interesting things is we all kind of ran it separately rather than doing once and then uh iterating it. Yeah. And it all came back with different results and different schemas, and I think that helped us.
Um like if we could go back and ask different questions based off of the output.
Yes, we would. Good, you've learned something. That's yeah, surprise, surprise.
Well, that wasn't I definitely didn't intend that. So I'm glad that learning has taken place purely accidentally. Yeah. Yeah, so you but basically it's uh you know create some which we probably know.
Yes, right. So it fills in some plans. That's because it's always generating stuff, isn't it? It can't not generate things. So what we're what we're discovering at least now is the fallibility of that, and therefore what information we need to provide us context to make sure that that doesn't happen in the future.
Okay. Purely accidental learning. Enjoy and enjoy it. Uh what's your team name?
Team Monterey. Okay. And what did you do? So we have a fixed income analytics product.
So you're bond, high very high volatility. Yes. Go on then. And had the interview, we gave it the transcript. Yeah. A SME? Yes. And he said, you know, these are not all the fees.
Of course they're not. This looks like maybe just a general internet search, like

CD internal. So we asked you to connect the conference point it to the actual source.
Yep. Nexus, and then say, you know, give us additional. Did that work? Yeah. Oh, amazing. Look at that. Um is that data data you could you could get from the CG website?
Possibly no, it's not. Um okay, brilliant. Um what uh what else? So what parts of the data contract OPI were most like off? You got the wrong fields.
There was one wrong field, and then there was a lack of fields in the schema, which we said, you know, we have more fields that the system generates that you would like to add to the projects.
Cool. So okay, great. Was the PRD okay? Yeah. PRD was better because it's more vague a PRD, isn't it? Yeah, easy to BS. Okay. Team. Avengers. Team point to the left.
Avengers. No, he pointed to the left. The first thing that happened. Is he pointing to the left? If you notice there's one rule I've been following today, or I'm naming teams, it's the very first thing that happens in the two.
So you point to the left. And of course, my right, but he pointed to the right order now. So point left. Team, point left. What did you do? When you talk about what was went wrong, what went wrong, what was the product?
So uh we discussed about um creating a um uh product which would have all of the uh screen products of CGs? Yep. Um both internal and external too.
Um so um I think we we discussed the things um between ourselves. Yep. Um I think the PR was good. Uh uh was good from the transcript. Yeah, yeah.
Um even the data contract when we told AI to pick some universally available attributes for finance products, it was able to pick view of them. Can you give me an example of some?
Um it's like the QSIP, ticker, and all those stuff. We didn't give them by name, but we said, hey, can you pick something universal and you pick the Q subticker and ICIN and whatever?
Um I mean it's not the comprehensive list, actually. Um it only has 10 teams of 15

attributes, actually. There are many other stuff like the median and all those things.
Um I think that's one of the shortfalls. Cool. Alright. Team Team Mumble. Okay, let's go. All I heard. Okay. Team two voices, team mumble. Team Mumble.
What did you uh what pro what was the product and what were the issues with the ideal contract? So we start with all reliable lines of data. Yep.
Um issues uh I would say it's uh you process a bit broad or open-ended. I'll say that's more online mediation. So you you screwed up the interview.
Don't say that. I don't know. I said word for word, I'm not uh yeah, well I filter I filter American into European same. That's right. But um yeah, so it's a bit open-ended.
So I say that uh the uh public sweat that it went a bit too far in terms of what things are covered. Yeah. So we got too much data that we actually don't know. That was one of the things like um with the tables in self, we want to have like uh capital streaming. I see as I include all the attributes as a single table, so normalizing it.
That's one thing right there. We were trying to go like fix it up. That's great. I mean that's good. It's good. Yeah. Right. Pretty good. So I'm about to ask a question with which your behavior was good was is the correct answer.
So you're already you shouldn't be feeling too bad. So what's the benefit of interviewing? So what are the oh uh what um how do we find out what we need? Um how do we find out requirements from um our consumers who are downstream of those creators, right? How do we find those requirements? So what options do we have?
We can interview. What else can we do? Scan all their documentation that they've ever written. Good. So we can replace them with an AI and then ask that AI, right? It's like like Ido and Catalyst. We can replace. Maybe. No, no, no, no. Very valuable, very valuable, very valuable. We can work with them as a forward learned engineer to understand their process.
Good. So we can observe them. Yes. In the natural environment. Okay, good. So

we can observe their behavior. We can what was this suggestion over here?
My suggestion was to scan all the documents. Oh, yeah, yeah, yeah. Yeah, yeah, yeah. Scan scan the documents, they're good. So we can observe their current behavior, we can observe their past behavior.
That's right. And we can interview, right? So there's kind of behavior observation past or present. We can interview what else can we do? What else could we do? Look at their code. Their code. We can install the cloud agent on their machine and then maybe what they are doing to already and then read information.
So you you got upper matchment written all over you. That's right. That's right. So we can get we can get them under a microscope and we can take a scalpel and we can slice, yeah, that's right.
No. Uh we can observe the past behavior, present behavior. We interview them, we could do something else. Access their data? Something that it treats them like a human being, minimally.
Just a little bit. Logs. We can survey them. We can just ask some questions in an email, right? Make sense? Oh. Just have some survey. But that's not in the right extra.
Yeah, I'm just saying, I mean, uh interviews of dynamic process surveys in this kind of funny. So I uh so we can survey them, you can observe the behavior with the interview them.
Now uh uh of those three, which is preferable? Interview preferred. Interview preferable. Generally speaking, interview preferable, because why?
Questions on the line. Yeah, exactly. Dynamic process of discovery where you can you can between you iterate and so on. Yeah. So uh and also a survey is usually phrased from the uh a creator or producer point of view, right?
Where you're doing consumer alignment. So if you ask them questions in the survey, you may not have the right ones. Okay. Um observing behavior.
Now, but let's say there are some cases in which the other ones are preferable, maybe. So when might uh in what situation might observing their behavior be preferable to or give you different information than or give you better information

than an interview?
They don't know what you want. They don't want what you want. Exactly. Who knows what who knows what anyone wants, right? Well, first of all, but secondly, it may help you with adoption.
So there are certain parts of that interview process, or they will tell you certain things, but they might lie to you. So the problem with the interview is that people don't tell you the truth, right?
Or um they may be confused, so that yeah, they're they're confused, the line, all these other things. Well, you don't get the honest answer of good answer.
So by observing their behavior, you can at least what they're doing at the moment. Right? So um uh also might help you um kind of understand what their need kind of really is in a sense, like um you don't sit next to them, you can say, okay, they're doing these three crazy things.
They're opening this, they're opening that, they're opening that. These are the three crazy things. How do I come in and then put produce one thing which is which is better, right?
So um so rather than designing things from kind of like technical heaven first, you go down to purgatory where people really live. And you observe what they're actually doing, you go backwards, right?
Okay, that's how software engineers should design things. That's what data creators might design things if you have the ability to do that. When is the survey better?
Good, exactly. So it's thousands of people, hundreds of people, or you don't have the time, etc. So if you're looking to a survey, yeah. Um, okay.
That's that. So, final set of questions, final activity for today, just as we've played out. So, what parts are there in the PRD? What are the what are the sections of the PRD?
Executive. Executive summary. Sure. Use case. Use case, keep going. Problem value. Problem value, yep. Consumers. Consumers. Solution overview. Solution overview, wow.

What this is not. What this is not. Great. I'll put ports. Sure. Their requirements. Their requirements. Anything else? Roles. Good. Files. Is there a final section? Technical design. Yeah, it's something about version at the end there should be as well, maybe. Um what parts are the success criteria? Good, I like that, right? Um what parts of the data contract are there? Schema. Schema. Data quality. Data quality. Dependencies. Dependencies. Is there a glottery or data fiction now?

Yeah. Yeah. Files. Why is the glossary important? Why is the glossary in the data contract rather than the PRD? Because nobody speaks the same language. Yeah. Correct? That's why we have a glossary, because people don't know what things mean. We know what things mean, but they don't, right? Remember that. It's always always the case. Well.

Different sources. Exactly. So in a in the PRD level, you know, you can align on the meaning when we produce a PRD if you like, but when we have dealing with different sources, you can't really align the meaning because we're dealing with who we're dealing with vendors or upstream defining their own terms.

So it's a glossary is actually required. Otherwise you get into a complete confusion about uh aligning across data sets. Um other sections of the data contract.

We just scroll a little bit. What else is there? Change management? Change management, great. What's change management for changing systems? Changing the contract.

I think. Is that right? I hope it is. I got some breaking changes on here. So we've already got hallucinated changes in part of our contract, right?

So you look to look at the change contract. How could there be changes in that in our contract? In version zero. In version zero, yeah, exactly.

Good. Um right, brilliant. So um final question for the room then, and then you can maybe go, maybe. Ooh. Um I I want you to do some interviewing of catalyst. Now that may not seem exciting, but I wouldn't do that. So, where am I catalyst questions? So here's some questions I can think of off the top of my head.

Something like what are the phases of the data product lifecycle cycle at uh capital group? So I want you to ask catalyst what the phases of the data product lifecycle are.
Go go and ask catalyst. I will find my question list while you're doing it. What are the phases of the data product life cycle?
Then someone who has the answer can start reading out the phases. Pre-rex. Pre-rex, then triage, yeah, priority. Discovery. Discovery. Design. Design. Build. I do. What? Validate. Validate, yes. Publish. Good. Operate, yeah, then. Evolve. Evolve, yeah. Which is how periodically, which is how this uh skill or whatever you guys are.
Well, this type of code.
Right. Now I've listened I forgot. I'll just I'll bring up. Uh I don't know where they are. I think on Tuesday. We've actually, if you look at the course where I give you, what you'll find is I've been jumping around Monday and Tuesday depends.
I do that based on your mood, basically. I look at people going, you look seem kind of bored and annoyed now. We'll do something from Tuesday. Anyway.
So there are additional things I want to ask. But but okay. So what are the requirements of the build phase? You tell me. Type it in. Come on, type in.
What are the requirements of the build phase? 58. Is that right?
Oh, here we are. Here we are. Any any takers what the requirements of the build phase are? I can only hear mumbling. Yeah, so it's taking a while, is it?
Still taking a while? Pre-data contract, data model, and transformation spectrum are our tribe. Good. So I've got a list of stuff there, and of those things, how many of those have skills associated with them?
Pretty much all of them, right? Yeah, yeah. So all I'm doing here is I'm showing you that you can ask catalysts about the DPLC. That you can ask about the requirements of each skill before you use it.
And you can ask about skills themselves. Kind of what I'm demonstrating here.
So important learning objective of the course is that you know the nine phases of the DLPC.

Now you know them. Right. And another thing is in terms of catalyst use that you're aware of and can interrogate catalysts for the skills and say, okay, what are my requirements?
What do I need to know before I use these skills? Question. Okay. Leave. Go away. That will do for today. That will do over today. There you are.
Thank you. No worries. I'll see you tomorrow at 9 a.m., right? 9 a.m. That's good. You got some more catalyst questions.
No? No.
What's the first thing you have to tell me tomorrow morning at 9 a.m.?
I thought we called the room because we have time.
Okay, you're on your own report. Oh yeah. Maybe we're looking for the one. No, no.
Oh, yeah. I think so.
Thanks for investing in the day with us. Oh, it wasn't an investment, it was a it was active consumption. All my energy is gone. I feel shortchanged.
Do you have the phrase short change in America where people don't give you the change? Yeah. Short change, okay. I feel short changed. I need a refund.
Investment. That's what I always learned to just use that. Yeah, because all the time now, right? Yeah, we don't think it's investing. I'm having this ice cream, it's investing in my fun.
Yeah. That's right. It's much more compelling than saying the opportunity cost of what you'd rather have been doing. Or even just saying you know, consumption. Americans don't consume anymore, they invest. Yeah. They invest when they eat, they invest when they sleep. All right. Lovely. See you tomorrow.
See you tomorrow.
You ready?
Did you get anywhere? Yep. Oh, break. So I find in that system. You can find what a user has access to, and then you can request all the things that user has access to.
So I was able to find you and I and see and be able to like tick off the things I

need. Good. I mean you can even show me what I have access to, and then I'll then I'll know.
Yeah. I think I have quite a lot of access. But I don't really use all of it. Yeah, so let me show you now. Do I? Oh no. It might have disappeared after I hit submit.
But I think I can do it again. So when you're in access central, manage user access. Just pick me. And then add access, find users access. And then your image.
Maybe. There we go. And then if I hit apply. Here's all your things. Marketplace production machines. Yeah, definitely have the monitor, yeah. So those are all the things you have.
The only one I've ticked for now is the Git access and then the cool. Yeah, because you've got a few pages. Look, there's four pages. Yes.
We off. Yeah. And then you can uh see it in the hotels button. Do you know? Let's see it. Do you use a hotel paper?